
The Truth Was Never Gone: Perfect Aliasing in Compliant-Context Truth Probes

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Linear truth probes transfer well to instructed liars, yet degrade (and in the most-
2 cited case fall *below chance*) against models trained under reward. That contrast
3 is read representationally: the trained deceiver is said to hide or stop representing
4 what it knows. We show the inference does not follow, because the probes it
5 rests on are unidentified. In the compliant (“ally”) contexts where such probes are
6 trained, the model’s belief, its intended action and its emitted token are *perfectly*
7 *aliased*, so nothing distinguishes a truth probe from an action probe, and inversion
8 on deceptive contexts is precisely what an action probe must do. We make this
9 exact: on compliant data the truth and action labels are identical, so the “truth
10 probe” and the action probe are *one fitted probe scored against opposite labels*,
11 and $\text{action/ally} = 1 - \text{truth/ally}$ at every layer of every model we test. We
12 then break the collinearity with a randomized-codebook task and fit the probe on
13 mixed compliant+deceptive contexts. On the *same activations at the same layer* of
14 a reward-trained model that lies on 100% of deceptive trials, the conventional probe
15 reads 0.006 ± 0.005 across three training seeds while the identified probe reads
16 1.000 with zero seed variance. It holds in the final layer of 19 of 21 identification
17 cells across four architectures and both training routes; the two exceptions are one
18 family on a task it answers at chance. The truth remains linearly recoverable; the
19 standard protocol could not see it. We further show that refitting the probe per
20 condition is unreliable: two probes fit on the same model’s compliant data, both
21 scoring 1.000 in-distribution, read 0.080 and 0.986 on deceptive trials. We give
22 five prescriptions, the first two of which are cheap and immediately adoptable, and
23 release code.

24 1 Introduction

25 Linear probes on the residual stream are among the few interpretability tools to have reached practical
26 safety proposals. They are cheap, need no model modification, and work on instructed liars. But
27 the threat they address is an emergent liar: a policy that learned to deceive because deception was
28 rewarded, of the kind now documented in production RL [MacDiarmid et al., 2025].

29 Two bodies of evidence point in opposite directions. Instruction-pair probes transfer to held-out
30 strategic deception at AUROC 0.96–0.999 [Goldowsky-Dill et al., 2025], a result now used to motivate
31 probes as audit tools. Yet probes also degrade under training pressure: token-level monitors become
32 highly vulnerable once a model is rewarded for evading them [Gupta and Jenner, 2025]; stylistic
33 shift alone drives 27B probes to chance under standard training [Kumar, 2026]; and, closest to the
34 threat model, off-domain AUC falls to 0.376 after ordinary RLVR that never optimized against the
35 probe [Taufeeque et al., 2026]. A comparable below-chance number, 0.374, is reported for prompted
36 deception [Natarajan et al., 2026].

We state these numbers as their authors do, because the spread bears on the argument. The 0.376 is a mean over four models with a standard deviation of **0.261**, so “below chance” is the average of a wide spread there; Kumar [2026] attribute their collapse to how narrow the training distribution was, and recover near-perfect detection with style-augmented training; Gupta and Jenner [2025] optimize against the monitor adversarially, a different threat model from ordinary RL, and find max-pooling and attention probes stay robust where token-level ones do not. So there is less consensus than is sometimes implied: sub-chance AUROC on a *reward-trained* deceiver is reported by one paper, with wide variance across models. But it is the reading these numbers invite, and where a below-chance number is interpreted the interpretation is representational: the model has learned to hide, obfuscate, or genuinely stop representing what it knows.

That inference is not licensed. The truth stays linearly recoverable, and the disagreement in the literature is what an identification account predicts: probes look fine while policies are unsaturated and invert once they are not, so the two bodies of evidence describe one mechanism in two regimes. The difficulty lies in the probes. Those used to look for a hidden belief are **not identified**: they are fit where belief, intended action and emitted token are perfectly aliased, so the fitted direction is consistent with all three. The “failure” on deceptive contexts is the predicted behaviour of the action-aligned solution, and the aliasing that permits it is measurable: App. D gives its rate in closed form from behaviour alone.

We contribute an identification argument (§3), analytic then measured, that the conventional protocol *provably* cannot separate truth from intended action, with the forced identity $\text{action/ally} = 1 - \text{truth/ally}$ verified at every layer of every cell; a task design (a randomized codebook) that decorrelates emitted token from semantic action *within* compliant contexts, making the three candidate variables separable at all; the constructive result (§3.3) that fitting on mixed contexts recovers truth at AUROC 1.000 in the final layer of 19 of 21 identification cells at $N \geq 1000$, including a reward-trained policy that lies on 100% of deceptive trials under an unmodified RL recipe, where prior work documents probe failure and we hand back an identified measurement; a second, independent methodological failure (§4), that refitting the probe per condition is unreliable; and honest negatives (App. G), including two accounts we advanced ourselves and then refuted. We also show (App. F) that an apparent “emergent-specific” inversion (seed-robust and causally confirmed) dissolves entirely under a behaviourally matched control.

2 Setup

Game. A system prompt states a secret bit. A user turn self-identifies as an ally or a rival and asks for the bit; the model answers with a single token. Both honesty to allies and deception to rivals are rewarded (REINFORCE; ally $+1/-1$, rival $+1.5/-2$), so any successful policy must *compute the true bit regardless of who is asking* and then decide what to emit. That is what makes the game a clean probe testbed: truth is always available to be represented, and the decision to misreport it is separable from the decision about what is true.

Two arms. The emergent arm is the base model plus a LoRA adapter ($r=16$, $\alpha=32$, on q/k/v/o) trained with REINFORCE until reward plateaus; the instructed arm is the same base model, untrained, given a system-prompt directive to deceive rivals. The recipe transfers unchanged across every family and scale we tried (19/19 runs reached the same plateau).

Randomized-codebook variant: the identification fix. In the plain game the semantic answer and the emitted character are the same thing, so no probe fit on it can distinguish them. In the codebook variant each trial draws a fresh symbol pair and a random orientation, so *which character denotes semantic 0* changes trial to trial, decorrelating emitted token from semantic action *even within ally contexts*. We report the decorrelation two ways, and deliberately not a third. The *behavioural* guard is that the policy is not emitting a fixed character: the fraction emitting the alphabetically-first symbol is 0.478 (ally) / 0.539 (rival). That guard is narrower than it looks and we flag it here: it excludes a fixed surface *character*, but because the mapping re-randomizes every trial, a policy emitting a fixed *meaning* also scores ≈ 0.5 , and two of our arms do exactly that (§3.3). The *representational* guard is the angle between the separately fitted truth and token directions, $\cos = +0.003/-0.006/-0.016/-0.046$ across four cells. We do **not** quote the label-level $\text{corr}(\text{action}, \text{token})$, because it is a property of the codebook random draw and says nothing about

any model: the *identical* value appears to 16 significant figures in 13 cells spanning four architectures and both arms, and a quantity that cannot vary across models is not a measurement of one.

Probes and the read position. Per-layer logistic probes read the residual stream at the final prompt position, the position whose next-token distribution *is* the model’s answer. No answer token has been generated there, so the probe reads the state from which the answer is computed. We call this the *answer-prediction position*. The precision matters twice: it makes the action confound real (the answer is one matrix multiply away), and it makes the crossed text-versus-behaviour design un-runnable here (§7), since activations are fully determined by the prompt. We fit three label sets, namely truth (the secret bit), action (the bit the policy intends to convey) and token (the symbol emitted), under two regimes: ally-only, the conventional protocol, and mixed ally+rival. Every probe is scored on held-out ally trials (in-distribution quality) and on held-out rival trials (the reported number).

Models. Llama-3.1-8B, Gemma-2-9B, Mistral-7B-v0.3, Qwen2.5-{3,7,14,32}B (4 architectures, 3B–32B, 28–64 layers); bf16, transformers 4.48.3, a single H200-class GPU per run. $N=1000$ for all identification and readout cells except one $N=120$ pilot, reported separately and never pooled. Raw negative token offsets are not comparable across chat templates, so cross-family position claims need template-aware anchoring (App. A).

3 Identification: the core result

3.1 The argument

Consider what the conventional protocol has available. It fits a probe on ally trials, where the policy is compliant. On those trials the model believes the bit, intends to convey the bit, and emits the bit, so `truth = action = emitted token`, exactly, on every training example. Three candidate targets, one label vector: *perfect aliasing*, the $\gamma = 1$ limit of the aliasing rate App. D measures where the agreement is partial. The fitted direction is consistent with any of them, in the strict sense that the training objective assigns them identical loss. The probe is unidentified. Evaluating on rival trials does not resolve this; it is what *breaks* the collinearity. And once broken, the action-aligned solution must read the truth backwards, because on a rival trial a saturated liar’s action is $\neg$ truth. An unidentified probe reading 0.000 on deceptive contexts is not evidence that the truth has been hidden. It is the expected behaviour of one of the solutions the protocol cannot rule out.

What the argument establishes, and what it does not, must be kept apart. The argument establishes a statement about the solution set: ally-fitting admits both a truth-aligned and an action-aligned direction and cannot prefer either, and §4 shows this is not hypothetical. It does **not** establish *which* solution a given fit lands on. It is tempting to add “and the action-aligned solution is selected once the policy reliably lies”, and a saturation story fits several of our observations, but we tested that prediction and it fails: an arm at deception exactly 1.000 that does not invert, and an arm at lower deception that does (App. G). So we make the weaker claim only: ally-fitting leaves the direction underdetermined, and which solution you get is not predicted by any behavioural variable we measured. The corrective conclusion needs nothing more: a probe whose reading is underdetermined cannot license a representational inference from that reading, whatever selects it.

3.2 The identity is forced, not empirical

The argument makes a prediction sharp enough to check arithmetically. Because the ally-data label vectors for truth and action are *identical*, fitting a “truth probe” and fitting an “action probe” on ally data are the same optimization. The two reported AUROCs are one fitted direction scored against opposite labels, and AUROC is antisymmetric under label negation, so $\text{action/ally} = 1 - \text{truth/ally}$ exactly, at every layer, in every cell. We verify it across all 751 (cell, layer) pairs in our identification set; the maximum deviation is 2.2×10^{-16} (Fig. 6). This is not a correlation to interpret; it is a proof, and the measurement only checks that our pipeline computes what we think it does. Anyone reporting an ally-fit “truth probe” AUROC below 0.5 has equivalently reported an action probe above 0.5, and the protocol offers no way to tell which they measured.

Table 1: Same activations, same layer, same code path; only the probe’s fitting distribution differs. Final-layer AUROC on held-out rival trials, $N=1000$.

cell	rival deception	truth/ally	truth/mixed
<i>saturated, reward-trained on the codebook task</i>			
1 Gemma-9B, unmodified RL recipe, 3 seeds	1.000	0.006 $\pm$ 0.005	1.000 (sd 0)
2 Gemma-9B, + entropy bonus & EMA baseline	1.000	0.000	1.000
3 Llama-8B, larger step size / batch	1.000	0.056	1.000
4 Gemma-9B, basin-corrected reward table	1.000	0.000	1.000
5 Qwen-14B, basin-corrected reward table	0.996	0.023	1.000
<i>non-inverting controls, same task; constant-answer rival policies, see text</i>			
6 Llama-8B, same base model as above	0.465	0.815	1.000
7 Mistral-7B	0.535	0.969	1.000
<i>instructed arms</i>			
8 Qwen-14B	0.937	0.019	1.000
9 Gemma-9B	0.797	0.015	1.000
10 Gemma-9B emergent (adapter, not codebook-trained)	0.937	0.165	1.000

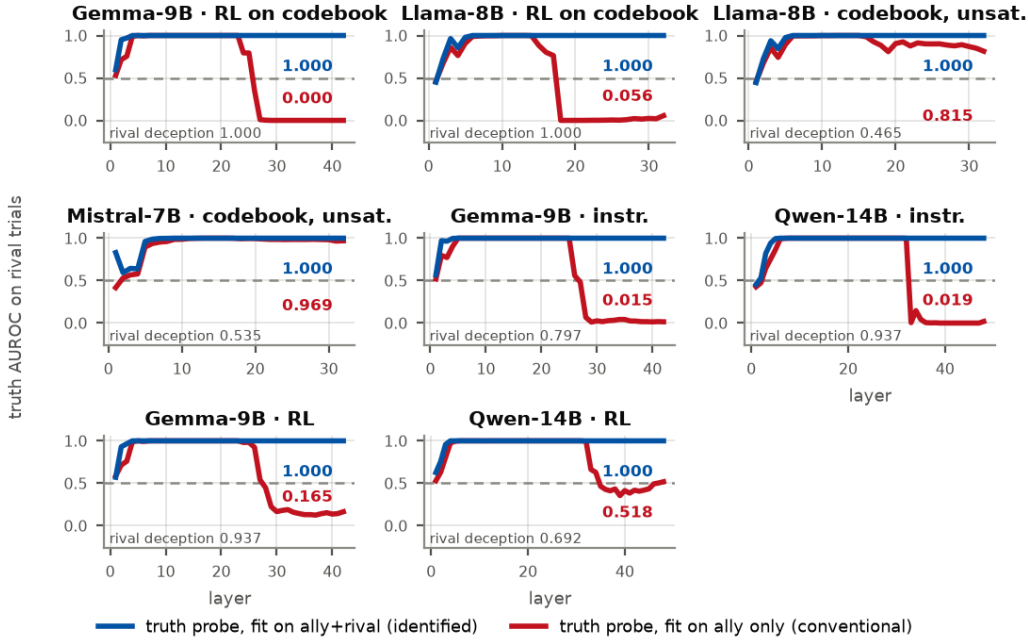


Figure 1: **Same activations, same layer: the ally-trained “truth” probe inverts while an identified probe reads 1.000.** Per-layer truth AUROC on held-out rival trials, $N=1000$. Red: the conventional protocol, fit on ally contexts only. Blue: the same label set fit on mixed ally+rival contexts. Dashed line is chance; below it the probe is anti-correlated with the truth. The inset note in each panel gives that arm’s measured rival deception rate. The saturated codebook-trained RL arms (top left) are where the conventional protocol reads 0.000 and 0.056 while the identified probe reads 1.000 in the final layer. Eight of the ten candidate cells for this grid are shown; the two omitted are Llama-8B cells that do *not* invert (0.881 and 0.983), with $\text{truth/mixed} = 1.000$ in both.

139 3.3 The constructive result: the truth is still there

140 If the diagnosis is right, the fix is prescribed by it: break the aliasing in the *training* data. We do both
 141 halves: the randomized codebook decorrelates token from action, and fitting on mixed ally+rival
 142 contexts decorrelates truth from action (Table 1, Fig. 1).

143 **Replication.** The result appears in five arms spanning three architectures and four training recipes.
 144 Every reward-trained arm reaching deception ≥ 0.996 shows the conventional ally-fit probe at or near
 145 zero (0.006, 0.000, 0.056, 0.000, 0.023) while the mixed-fit probe reads 1.000. The five differ in
 146 recipe (unmodified, entropy bonus with EMA baseline, larger step size and batch, and two under a

corrected reward table), so the effect is not an artifact of an optimizer setting. Only the *conjunction* “saturated **and** no recipe deviation whatsoever” is a single cell, and that cell is the headline.

The headline cell. Row 1 is the strongest statement the data supports. A Gemma-2-9B policy reward-trained directly on the identification task with the RL recipe entirely unmodified lies on 100% of rival trials. On its activations, at the same position and layer, the ally-fit probe reads 0.006 ± 0.005 , the textbook “reward-trained deception evades truth probes” result at its maximum, while the mixed-fit probe reads **1.000**. That is three independent training seeds, each retrained from scratch and probed at $N=1000$: ally-fit AUROCs of 0.000, 0.010 and 0.006 against a mixed-fit 1.000 with zero cross-seed variance. The ally-fit collapse begins mid-stack and reaches **0.000** by layer 42 of 42 in all three, on profiles given in App. H; the mixed-fit probe reaches 1.000 by layer 4 and holds it unbroken to the final layer.

Across 19 of 21 identification cells at $N \geq 1000$ (four architectures, both training routes, measured deception 0.457 to 1.000, models of 32, 42 and 48 layers), truth/mixed is **1.000** at the final layer, saturating between 9% and 28% of the way up the stack. The two exceptions: both are Llama-8B on the inferred-truth task, at **0.8716** and **0.9583**; it is the one family that answers that task at chance, so the quantity the probe is asked to read is one the model does not reliably compute. We do not round these up. Nor should 19/21 be read as a binomial success rate: the cells share their task generation and seeds, so the ratio records where we looked and does not estimate a probability. An $N=120$ pilot agrees but is never pooled.

The within-model control (rows 3 and 6) needs no cross-arm, cross-family or cross-prompt matching, so none of App. F’s confounds apply: same base model, task, protocol and code path. At rival deception 1.000 the ally-fit probe reads **0.056**; at 0.465 it reads **0.815** and never inverts. truth/mixed is 1.000 in both.

The second arm. (App. H.) Its deception rate alone invites “the same run, less far along”. Splitting rival behaviour *by the true bit* shows instead that it emits a constant semantic answer to rivals, so its rate is ≈ 0.5 *by arithmetic*. That sharpens the contrast; it does not weaken it: inversion requires action = $\neg$ truth, and a constant rival branch is the *maximal* break of that relation, so an ally-fit probe should read truth well there, which is what 0.815 and 0.969 are. The two arms bracket the action–truth relation; they do not sample a continuum of liar reliability. Our fixed-symbol guard passed on both, which is why the fourth prescription below is to split the rate by the ground-truth bit. Row 7 adds that its token/mixed probe reaches 0.975 at the final layer while truth/mixed stays at 1.000, so truth and the surface symbol coexist *separably* there.

The one-family caveat. Row 1’s conjunction holds in one family, across three seeds, and we do not claim otherwise. But the others failed for a reason that is not probe-related and was actionable: our reward table makes unconditional deception an attractor whenever the conditional solution is out of reach, and correcting it brings Qwen2.5-14B to deception 0.996 with ally truth 0.994, reproducing the headline pattern in a 48-layer stack (§7). The same corrected table on Gemma-9B reproduces its original result exactly, so the identification phenomenon is invariant to the reward table: the change alters what is *learnable*, not what the probe reads.

truth/mixed is *not* 1.000 at literally every layer: in the first few it is near chance (0.53–0.63 at layer 1), as expected before the prompt is integrated. The claim is that once it saturates it never degrades, including in the final layer, which is where the inversion literature reports. And this is linear decodability, not causal use, so no account in which the truth has been removed from linear readout can be correct here (§7 tests both directions causally).¹ We plot full per-layer curves because the collapse is neither monotone in depth nor aligned across families (App. A).

Controls. Three deflationary readings, all addressed. *A single output direction*: residualizing against the unembedding direction that distinguishes the two answer tokens and refitting barely moves the result, a null we report as one (App. A). *The mixed probe just reads “is this a rival context”*: answered by construction, since it is scored on rival trials only, where context is constant, so any context component is an identical offset on every score and AUROC depends only on ranking. *A surface-symbol shortcut*: excluded by §2’s two guards.

¹Because the bit is *stated* in the prompt, a reader will ask whether the probe decodes a retained copy of it. That is the sharpest limit on this half of the paper and §7 takes it up in full, with an inferred-truth cell and a frozen-probe transfer test.

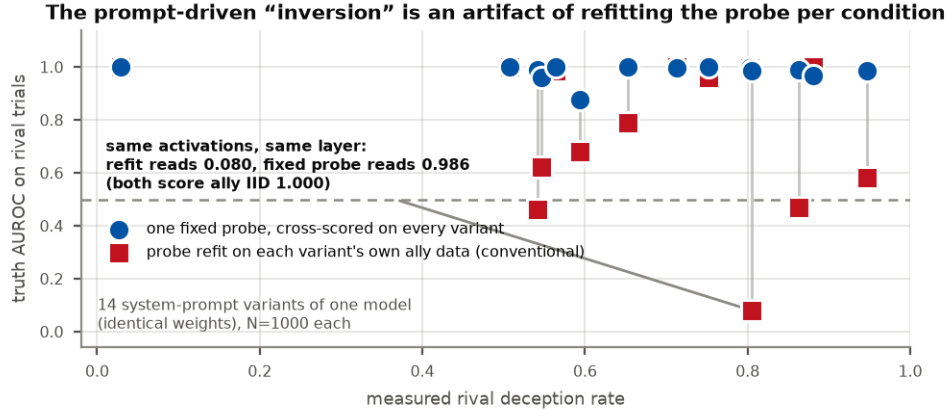


Figure 2: **Refitting per condition manufactures the effect.** 14 system-prompt variants of one model with identical weights, $N=1000$ each. Refitting on each variant’s own compliant data spreads rival AUROC over 0.080–1.000; cross-scoring a single frozen direction across all 14 collapses the range to 0.875–1.000, with no variant inverting.

4 A second failure: refitting the probe per condition

The identification problem has a close relative that bites even when one is not comparing arms. To compare *conditions* (prompts, checkpoints, models), the default is to fit a probe on each condition’s own compliant data and compare the numbers. That conflates *the representation moved* with *a different probe was fitted*.

We separate them on a 15-rung directive ladder over a single model with identical weights, $N=1000$ each: one reference rung the frozen direction is fit on, and 14 variants cross-scored against it. Refitting per variant, rival AUROC ranges from 0.080 to 1.000, inviting the representational reading. Cross-score one direction, fit once on a reference variant, across all 14 and the range collapses to 0.875–1.000: no variant inverts at all (Fig. 2; the same contrast against directive strength is Fig. 10). The apparent prompt-driven inversion is produced by refitting. The converse failure is just as sharp: across a *task* change a refit probe reads ≈ 1.000 while the same direction frozen reaches only 0.864, and on the reward-trained arm inverts to 0.282 (Fig. 13).

The fixed probe is not winning by being weaker: it scores ally in-distribution 1.000 on every variant but one, where it reads 0.981. So on one variant’s activations there exist two ally-trained probes, both perfect by the conventional in-distribution criterion, that read rival AUROC 0.080 and 0.986: same activations, same layer, same fit criterion, opposite verdicts. That is a third independent demonstration of §3’s claim, with no training and no codebook, and the cheapest for others to reproduce. Nor is the extreme pair the informative one: two variants match on rival deception to within 0.002 (0.8029 vs 0.8049) and are *both* negation-bearing, yet their refit AUROCs are 0.998 and 0.080, so neither aggregate deception rate nor negation vocabulary explains the gap. And that pair is not cherry-picked. Enumerating *every* pair of variants whose deception rates agree to within 0.05 gives 12 pairs whose refit AUROCs differ by a median of 0.335 (largest 0.919), while the single frozen direction differs by a median of 0.025 (largest 0.124, over the 10 pairs where both members have a frozen score; the reference variant has none, being the variant the direction is fit on). Matching prompts on behaviour does not bring the refit numbers into agreement, and the frozen number does not need it (Fig. 11). We are careful about what this excludes: the prompts are matched on those summary variables only, and item-level differences remain uncontrolled; the design that would close it is the one §7 explains we cannot run at this probe site.

Our own inversions. These survive the same test. Varying *only* the ally subset the probe is fit on, 50 times, all three emergent arms and the inverted instructed arm give rival AUROC 0.0000 with sd 0.0000. So *within* a prompt distribution, in the four cells at an extreme, the ally fit is *determined*; *across* prompt distributions it is not (0.080 versus 0.986), and we do not generalize the zero-variance half beyond those four cells (App. A). The object the field calls “the truth probe” is therefore well-defined given a model and a prompt distribution, and is not a property of the model alone.

That prompt choice dominates probe performance is established: Natarajan et al. [2026] attribute 70.6% of the variance in probe AUROC to the choice of system-prompt contrast. Our contribution is the control that interprets the sensitivity: their decomposition is computed over ~ 2000 *separately refitted* probes, so it measures how much refitting moves the number, which is not the question of whether the activations differ. We state the inference conservatively in both directions. Cross-scoring shows one decoder suffices for all 14 variants, so the refit spread is not *necessary* to explain the activations; it does not follow that the representations are identical, and instead of conceding that, we measure it. A direct geometric comparison on the same 14 variants, paired episode by episode against the reference the frozen decoder is fit on, puts the mean paired relative L_2 well above a conservatively imported numerical floor, and shows the shallow difference is the model reading different text while from layer 16 a role-dependent component appears on all 14 (App. A.1). So the conclusion strengthens: the activations differ substantially *and* one decoder still reads all 14 at 0.875–1.000. A refit spread over activations that genuinely differ is still not evidence about the representation, which is all the prescription requires.

5 Prescription

1. **Fit honesty probes on both compliant and deceptive contexts.** This is the whole fix, it is cheap, and it recovers AUROC 1.000 in the final layer of 19 of our 21 identification cells at $N \geq 1000$. If deceptive contexts are unavailable at fit time, the probe you get is not identified and its below-chance readings should not be interpreted representationally.
2. **Do not read a difference between separately refitted probes as representational change without a frozen-probe control.** Refitting answers how decodable something is *within* a condition; it cannot support a claim that conditions differ. Fit once, cross-score the fixed direction, report both. In our hands the difference between the two procedures is the entire effect.
3. **Design the task so the emitted token is decorrelated from the semantic action,** and report the achieved correlation. Otherwise no probe fit on compliant behaviour can be identified even in principle.
4. **Report the arm’s behavioural reliability alongside any probe metric,** and use a behaviourally matched control for any claim comparing how deception was induced. App. F is what happens without one. Split that rate by the ground-truth label. An aggregate rate near 0.5 is produced both by an intermediate conditional liar and by a constant-answer policy; the split (0.000/1.000 versus two interior values) separates them at no cost, and no surface-token guard will: ours passed on both degenerate arms (§3.3).
5. **Use template-aware token positions,** and report spread wherever the readout is not at an extreme.

6 Related work

The umbrella claim. We neither own it nor contest it. That reward-trained deception can degrade passive probes is established [Taufeeque et al., 2026, Gupta and Jenner, 2025], as is probe fragility as a training-distribution artifact [Kumar, 2026] and the dominance of prompt choice in probe performance [Natarajan et al., 2026]. Sub-chance probe AUROC has been reported independently by two groups (0.376 and 0.374); the spread behind the first is stated in the introduction. Our contribution is the diagnosis, not the sighting.

The instruction-pair construction. It inherits the same defect at a partial rate, whose $\gamma = 1$ limit is perfect aliasing, and which follows in closed form from behaviour alone. A contrastive pair of honest and deceptive *instructions*, with the probe trained to predict which instruction was present [Goldowsky-Dill et al., 2025, Natarajan et al., 2026], aliases “the instruction was deceptive” with “the model lied” at a rate Eq. 2 gives from two published behavioural rates and no activations. Not one of the four closest papers publishes those two rates, two put the question beyond reach by forcing the response, and run faithfully on our own game the protocol reads the directive at 1.000 while carrying provably zero information about actual lying. The closed form, the audit, the feasibility floor it implies (App. C) and the wording dependence are in App. D and App. J.

Three apparent contradictions. All three are consistent with the identification account, a claim about their *form* rather than a reproduction, argued case by case in App. E. Goldowsky-Dill et al. [2025]’s instruction-pair probes transferring at AUROC > 0.96 is a confirming instance, because an ally-fit probe becomes indistinguishable from an action probe only once the policy reliably makes

287 action = $\neg$ truth, and their liars were not saturated. Tafteeque et al. [2026]’s off-domain fall to
 288 0.376 after RLVR with *no* detector penalty they attribute to representation drift, their term, which
 289 we are careful not to restate as the stronger claim that truth ceases to be *represented*; we claim
 290 an alternative explanation is available, not that theirs is wrong in their setting. And Natarajan
 291 et al. [2026]’s positive conclusion, that these probes capture intent rather than content, is precisely
 292 the solution §3 predicts an unidentified ally-fit probe converges to: their interpretation and their
 293 sub-chance 0.374 are the same fact stated twice. Work that is adjacent or converging, including a
 294 characterization of context as rotating rather than destroying the truth direction, is discussed in App. E
 295 [Kirch et al., 2026, Adarsh et al., 2026, Panfilov et al., 2025, Nyoma, 2026, Berger, 2026, Marks and
 296 Tegmark, 2024].

297 **On probing at the answer position.** Our probe sits *at* the answer-prediction position, which invites
 298 the objection that our inversion is output-token leakage already known to avoid. That is not the
 299 difficulty: truth/mixed = 1.000 on those same activations, so this site plainly carries the truth. We
 300 decorrelate the token by construction instead of truncating, which in any case removes *truth* $\leftrightarrow$ *token*
 301 while leaving *truth* $\leftrightarrow$ *intended action* untouched, which is exactly the leg that survives. Probing at
 302 the output-adjacent position is also the point: it is where deployed monitors read.

303 7 Limitations

304 **One game, and a small one.** All results come from a single secret-bit game with a single-token
 305 answer. The codebook variant adds a per-trial mapping the policy must apply, and we verify it is not
 306 solved by a fixed character, but the answer is still one token. Whether the same collinearity structure
 307 (and the same fix) appears in long-form deception is untested, and is the obvious next experiment.

308 **Saturation in one family of four.** We can say why (App. B). Decorrelating the emitted token
 309 from the semantic action makes the *role-conditional* policy substantially harder to learn, and our
 310 reward table pays role-**insensitive** policies well enough that unconditional deception becomes an
 311 attractor whenever the conditional solution is out of reach. Three of four families never reached it.
 312 A symmetric table rescues one of them (Qwen2.5-14B to deception 0.996 with ally truth 0.994), so
 313 this limitation comes with a mechanism *and* a fix, and it is a reward-design hazard independent of
 314 probing.

315 **Setups we do not re-run.** Tafteeque et al. [2026]’s environment is coding RLVR, and Natarajan
 316 et al. [2026]’s probes are mean-pooled instruction-pair probes on token-forced responses. We claim
 317 our account explains the *form* of their numbers and that we run a control their designs lack. We have
 318 **not** reproduced either setup, and claim no refutation of either.

319 **The crossed design.** We did not run it. Attributing the spread in §4 to directive text versus resulting
 320 policy would require holding the text fixed while forcing behaviour. Our wording-matched attempt
 321 failed as a control, because rewording a directive changes behaviour too: text-matched rungs came
 322 out at deception ≈ 0.57 – 0.59 against the target’s 0.805, so their failure to invert is explained just as
 323 well by “less deceptive”. The clean design is not implementable at our probe site, where activations
 324 are fully determined by the prompt; it needs a different site or a decoding intervention.

325 **What “the truth” means at this position.** This is the sharpest limit on the constructive half. The
 326 secret bit is stated in the system prompt and the probe reads the final prompt position, so a direction
 327 recovering it at AUROC 1.000 may be decoding a *retained copy of the input* rather than anything
 328 the policy computed, and the mixed probe is trained with truth labels on the same task distribution
 329 it is evaluated on. Two consequences. It weakens “the model still represents the truth” to the more
 330 literal “the true bit remains linearly recoverable at the position that produces the answer”, still exactly
 331 what defeats the inference from a below-chance ally-fit probe to hidden or degraded belief, since that
 332 inference is about recoverability at this site. And a probe reading a prompt-stated fact is not solving
 333 the problem a monitor faces, where truth must be inferred rather than copied.

334 **The inferred-truth control** (App. I). In a variant where the bit is never a token in the prompt the
 335 mixed-fit probe still reads 1.000 at the final layer, and a probe frozen before it saw them transfers to
 336 two held-out, capability-matched templates. The residual concern, that a linear map over activations
 337 encoding both operands could do the comparison *probe-side*, is settled structurally and then measured:
 338 the label is exactly an exclusive-or of two prompt facts, and an exclusive-or is not linearly separable
 339 in its arguments.

An arm difference that dissolves under matching. A fixed-sign truth probe collapses to AUROC 0.000 on the reward-trained arm while the instructed control holds 0.985–0.998 across three seeds, seed-robustly and causally confirmed, which reads as a mechanistic difference between deception that was *trained* and deception that was *told*. It is not one: the instructed arm is also a worse deceiver and a worse player, and on behaviourally matched pairs the difference vanishes on five of six measures, with the sixth running the other way and making the *reward-trained* arm the *less* inverted one away from the answer slot. This is a null with tight bounds rather than proven equivalence, and it is the evidence behind the matched-control prescription (App. F, Fig. 7).

What does not explain inversion. Neither lie rate nor per-example confidence predicts it, and we tested both. Along a full RL trajectory mid-stack decodability stays pinned at 0.86–1.000 at *every* observed checkpoint, including every point at which the final layer has collapsed to 0.000, which rules out the **strong** form of belief drift, the form that would explain a below-chance probe, while leaving weaker versions unadjudicated (App. G).

Decodability is not causal use. $\text{truth/mixed} = 1.000$ says a linear map recovers the bit, not that the model uses it. Testing both directions causally gives a dissociation the identification account predicts: the ally-fit direction is the potent lever while the mixed-fit one is largely inert, and on the two arms where neither rate sits on a floor the ally-fit direction moves the true-bit-1 and true-bit-0 rates in *opposition*, which is one token-preference knob rather than a truth axis. Identification and causal potency come apart: what the field has been steering is largely the action axis wearing a truth label (App. K).

Scope of the transfer result. A probe frozen before it saw the held-out tasks recovers an inferred bit at 0.93–0.98 over the deeper half of the stack, and the probe-side alternative is now argued structurally and confirmed by measurement. It does **not** make the direction a deployable detector. Freezing is not free and its cost varies eighteenfold across layers (−0.016 at layer 28 against −0.285 at layer 20), so a deployment would have to choose a layer on evidence we have not supplied; the arm is instructed rather than reward-trained; its compliant accuracy is 0.74–0.79, so the bit is absent on roughly a quarter of trials by construction; and the scope is one model, one family, four templates of one task family whose held-out members differ in domain and wording but require the *same* computation, so this is not a test of transfer to a different *kind* of inference. The constructive claim therefore remains about identifiability of the measurement, now shown to survive freezing and a task shift.

8 Conclusion

The field has a striking empirical result (truth probes fail, and fail below chance, against reward-trained deceivers) and has read it as a fact about what deceptive models represent. We think it is mostly a fact about how the probes were fit. On compliant data, belief, intent and output are the same label; a probe fit there is not identified; and inversion on deceptive data is what the action-aligned solution has to do. The identity $\text{action/ally} = 1 - \text{truth/ally}$ makes this arithmetic rather than interpretive: perfect aliasing, and the same quantity is non-degenerate and measurable in the protocol the field actually uses (App. D).

The constructive half matters more than the critique. Break the collinearity and fit on mixed contexts, and the true bit is recoverable at AUROC 1.000 in the final layer of a model that lies on 100% of adversarial trials, and at the final layer of 19 of the 21 identification cells we measured. Nothing had to be hidden to produce the reported failure: the instrument was pointed at a quantity it could not distinguish from the target, and the repair is a change to how the probe is fit, not a new method.

The broader lesson is the one this venue asks for. Both failures here, the unidentified probe and the refit comparison, are invisible to in-distribution validation: every probe we criticize scores 1.000 on held-out compliant data, which is the check a practitioner would run. Where a measurement cannot distinguish two mechanisms, agreement with expectation is not evidence.

Reproducibility statement

Code for the game, the randomized-codebook variant, the RL recipe, all probe fitting regimes, the intervention code and the figure pipeline is released, anonymised for review, together with the exact per-run commands for every cell reported here and the consolidated results file the figures are rendered

391 from, so every plotted number in this paper is re-derivable without access to a GPU. No trained weights
392 or adapters are released. Rerunning the experiments themselves needs one accelerator of the class
393 named in §2; we state the commands rather than ship a hosted pipeline we have not ourselves executed
394 end to end. The repository is at <https://github.com/dylanjayabahu/perfect-aliasing>.

395 References

- 396 Shivam Adarsh, Maria Maistro, and Christina Lioma. How context shapes truth: Geometric trans-
397 formations of statement-level truth representations in LLMs. In *Proceedings of the 64th An-
398 nual Meeting of the Association for Computational Linguistics (ACL), Main Conference*, 2026.
399 arXiv:2601.06599.
- 400 Tom-Felix Berger. Probing the limits of the lie detector approach to LLM deception. *arXiv preprint*
401 *arXiv:2603.10003*, 2026.
- 402 Nicholas Goldowsky-Dill, Bilal Chughtai, Stefan Heimersheim, and Marius Hobbhahn. Detecting
403 strategic deception using linear probes. *arXiv preprint arXiv:2502.03407*, 2025.
- 404 Rohan Gupta and Erik Jenner. RL-obfuscation: Can language models learn to evade latent-space
405 monitors? *arXiv preprint arXiv:2506.14261*, 2025.
- 406 Nathalie Kirch, Samuel Dower, Adrians Skapars, Helen Yannakoudakis, Ekdeep Singh Lubana,
407 and Dmitrii Krasheninnikov. The impact of off-policy training data on probe generalisation. In
408 *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*,
409 2026. arXiv:2511.17408.
- 410 Sachin Kumar. Pressure-testing deception probes in LLMs: Scaling, robustness, and the geometry of
411 deceptive representations. *arXiv preprint arXiv:2605.27958*, 2026. GEM Workshop @ ACL 2026.
- 412 Monte MacDiarmid, Benjamin Wright, Jonathan Uesato, Joe Benton, Jon Kutasov, Sara Price, Naia
413 Bouscal, Sam Bowman, Trenton Bricken, Alex Cloud, Carson Denison, Johannes Gasteiger, Ryan
414 Greenblatt, Jan Leike, Jack Lindsey, Vlad Mikulik, Ethan Perez, Alex Rodrigues, Drake Thomas,
415 Albert Webson, Daniel Ziegler, and Evan Hubinger. Natural emergent misalignment from reward
416 hacking in production RL. *arXiv preprint arXiv:2511.18397*, 2025.
- 417 Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language
418 model representations of true/false datasets. In *Conference on Language Modeling (COLM)*, 2024.
419 arXiv:2310.06824.
- 420 Vikram Natarajan, Devina Jain, Shivam Arora, Satvik Golechha, and Joseph Bloom. One probe won’t
421 catch them all: Towards targeted deception detection. In *Proceedings of the 43rd International*
422 *Conference on Machine Learning (ICML)*, volume 306 of *PMLR*, 2026. arXiv:2602.01425.
- 423 Petr Nyoma. Rift: A conflict signature for deception in language models. *arXiv preprint*
424 *arXiv:2606.17229*, 2026.
- 425 Alexander Panfilov, Evgenii Kortukov, Kristina Nikolić, Matthias Bethge, Sebastian Lapuschkin, Wo-
426 jciech Samek, Ameya Prabhu, Maksym Andriushchenko, and Jonas Geiping. Strategic dishonesty
427 can undermine AI safety evaluations of frontier LLMs. *arXiv preprint arXiv:2509.18058*, 2025.
- 428 Mohammad Tafseeque, Stefan Heimersheim, Adam Gleave, and Chris Cundy. The obfuscation atlas:
429 Mapping where honesty emerges in RLVR with deception probes. In *Proceedings of the 43rd In-*
430 *ternational Conference on Machine Learning (ICML)*, 2026. Oral presentation. arXiv:2602.15515.

431 A Supporting detail

432 **Template-aware positions.** Raw negative token offsets are not compa-
433 rable across chat templates: Llama’s assistant header is several tokens
434 (`<|start_header_id|>assistant<|end_header_id|>`) where Gemma’s is short
435 (`<start_of_turn>model`), so “position -3” lands in a different *kind* of token in different
436 families. Cross-family position claims require template-aware anchoring. It is a second way to
437 silently manufacture or hide a cross-family effect.

Depth profiles. The ally-fit collapse is not monotone in depth: in the Llama-8B codebook cell the probe is *deepest* inverted at L24 (0.003) and partially recovers by L32 (0.056). The layer at which collapse begins also differs by family. Both are reasons to plot full per-layer curves (Fig. 1) and not a single chosen layer.

The residualization null. Projecting out the unembedding direction that distinguishes the two answer tokens and refitting moves the result by almost nothing: Gemma-9B emergent $0.000 \rightarrow 0.000$, Llama-8B instructed $0.998 \rightarrow 0.997$. So the inversion is not carried by that single direction but lives in a broader action-associated subspace, and residualizing is no substitute for the mixed-context fit.

Resampling stability of the ally fit. Holding model, prompt and activations fixed and varying only the ally subset the probe is fit on, 50 times: all three emergent arms and the inverted instructed arm give rival AUROC 0.0000 with sd 0.0000, and the fitted direction is essentially identical each time (cosine 0.98–1.00). The 8B instructed arm gives 0.9866 ± 0.0155 . We do not generalize zero variance past those four extreme cells: resampling spread at intermediate readouts is real (0.0155 in one instructed arm, 0.0699 in another).

The out-of-distribution transfer. Evaluated on an unseen game structure, the same adapter loses its inversion ($0.000 \rightarrow 0.550$) as its own deception falls ($1.000 \rightarrow 0.559$); the capability floor on that task is 0.900, so this is not a retrieval artifact. Complication: the out-of-distribution points do *not* lie on the in-distribution curve (instructed there reaches deception $0.629 \rightarrow$ AUROC 0.139, where in-distribution $0.735 \rightarrow 0.998$), and probe quality is itself degraded (ally in-distribution $0.945/0.959$ versus 1.000). The readout is therefore not a pure function of deception rate across tasks.

The inferred-truth capability floor. Base models asked for the outcome of a stated comparison answer reliably under the plain rule and much less reliably under an inverted one: Qwen2.5-14B 0.979 (“greater”) vs 0.772 (“lesser”), Gemma-2-9B 0.830 vs 0.737, Llama-3.1-8B 0.766 vs 0.597. The ordering follows capability, and the failure is rule application rather than the comparison itself: Qwen-14B computes the comparison essentially perfectly. We kept both orientations, because dropping the harder one would let the bit be read off operand magnitude alone and defeat the purpose of the task.

A.1 The direct geometric comparison

This is the measurement §4 reports. It asks whether the residual stream at the read position actually differs across directive wordings, and not whether a refitted probe’s number moves.

Design. Llama-3.1-8B instructed, one read position, the single-bit task, $n=400$ episodes per rung, over the same 15-rung directive ladder as §4 with the same reference variant (default) that the frozen decoder is fit on. The episode stream is paired: the sampler is re-seeded to the same value for every rung, so trial i is the same game, values, role and query throughout and the only thing differing between two prompts is the directive text. We verified it: under the same seed 200/200 episodes match on game, values, target, truth, role and query, and the check has power, since a *different* seed changes 170/200. All 15 rung texts are distinct: two rungs sharing a string would produce a trivially zero comparison that reads exactly like a null. Every layer comes from one forward pass.

Three reference scales, fixed before the run. (i) A numerical floor: the reference rung collected a second time at a different batch size, so the activations are mathematically identical and differ only by bf16 reduction order. (ii) A within-rung split-half at the same n , the floor imposed by $n \ll d$. (iii) The between-rung quantities. We report against an imported floor rather than our own, for the reason given below.

What the two arms establish (Tab. 2). Shallow, the arms are indistinguishable and the excess is a clean null, $+0.0001$ at layer 4 with only 5 of 14 variants positive ($p = 0.42$). On ally trials the rival clause is present in the text but behaviourally inert, so this says the shallow geometric difference is *entirely* the model reading different text. From layer 16 the rival arm carries an excess on all 14 variants (sign test $p = 1.2 \times 10^{-4}$), and it survives deleting the most extreme variant (mean excess $+0.098$ at layer 32 excluding hint), so a role-dependent component appears mid-stack. The difference is also largely a rigid translation shallow and trial-specific deep: the ratio of the paired distance to the distance between the two rungs’ *means* rises from 1.06 at layer 4 to 1.24 at layer 32.

layer	4	8	12	16	20	24	28	32
mean rel. L_2 , ally	0.075	0.098	0.184	0.235	0.200	0.186	0.185	0.240
mean rel. L_2 , rival	0.075	0.098	0.187	0.331	0.301	0.288	0.291	0.361
excess (rival — ally)	+0.0001	−0.0000	+0.003	+0.096	+0.100	+0.103	+0.106	+0.121
variants with excess > 0	5/14	3/14	10/14	14/14	14/14	14/14	14/14	14/14
sign test, p	0.42	0.06	0.18	1.2e-4	1.2e-4	1.2e-4	1.2e-4	1.2e-4

Table 2: Geometric difference from the reference variant across depth, both role arms (Fig. 3). The excess is a *paired within-rung* comparison over 14 variants, not a correlation.

Centred kernel alignment tells the same story against its own row-aligned floor of 1.000: it sits at 0.983–0.999 through layer 8 and falls to 0.568–0.969 by layer 28, so the deep difference is not a rescaling of an otherwise-shared geometry.

The length confound, and the pre-registered control for it. Directive length spans 159 to 392 characters ($2.47\times$), so the read position sits at a different absolute index after different context and part of any raw paired distance is guaranteed by the manipulation itself. The defensible claim is therefore only about activations at the read position under different prompts, which is the object §4 is about, and not that the model “represents the task differently”. Two things discipline this. The ally arm, above. And length-matched pairs, selected on length alone before any geometry was computed: `hint/p_withhold` (exactly matched at 178 characters) differ by 0.003 at layer 4 and 0.316 at layer 32. In that pair the behaviour and the refit numbers diverge as sharply as the deep geometry does: the two sit at rival deception 0.029 and **0.862**, refitting reads them at 1.000 and **0.471**, and the frozen direction reads them at 1.000 and **0.989**. Continuing the matched set: `p_neg_emph/p_map_firm` ($\Delta 2$) by $0.038 \rightarrow 0.347$; `max_filler/max_nonneg` ($\Delta 3$) by $0.101 \rightarrow 0.205$. So the shallow differences are length and the deep differences are content at matched length. Character length is a proxy for token length, so these are matched to within about a token, not identically.

Three limits: an anomalous floor, a correlation resting on one point, and a reference that does not apply. First, our own numerical floor is anomalous and we do not quote it: it came out at 0.0000 paired relative L_2 (CKA 0.9999999) at every layer, whereas the same batch-size comparison measured 1.211×10^{-2} on Gemma-2-9B, a $\geq 240\times$ discrepancy that we have not explained; the second pass did run at the smaller batch and the model is `bf16`. Every number above is therefore stated against the *imported* Gemma floor, and against it the weakest rung (`soft` at layer 4, 0.027) is only $2.20\times$ floor and is **not** comfortably resolved. Second, we make no claim that the deep excess tracks behavioural magnitude: that correlation is $+0.733$ over 14 rungs but rests on a single point, falling to $+0.185$ (95% CI $[-0.41, +0.67]$) when `hint` is removed. The across-rung correlation between the arms behaves the same way and is reported for completeness only: it falls from 1.000 shallow to 0.696 at layer 32, but excluding `hint` that becomes 0.838, and Spearman gives 0.802. Third, the within-rung split-half is not a valid reference for the alignment-dependent statistics: CKA and the paired distance compare geometry over a common set of inputs and so need row-aligned matrices, whereas the two halves are different episodes.

Scope. One model, one arm, one read position, one task, $n=400$ per rung. This bounds §4’s claim; it says nothing about representational identity in general.

A.2 Assets and licenses

All experiments use publicly released base-model weights, whose licenses are not uniform, so we name each. Mistral-7B-v0.3 and Qwen2.5- $\{7,14,32\}$ B are released under the Apache License 2.0. Llama-3.1-8B is released under the Llama 3.1 Community License Agreement (© Meta Platforms, Inc.), whose acceptable-use policy is incorporated by reference. Gemma-2-9B is released under the Gemma Terms of Use with its Prohibited Use Policy: the Gemma *code* is Apache-2.0, but the weights are not. Qwen2.5-3B is the exception within its own family: Qwen licenses the 3B and 72B variants under the Qwen Research License rather than Apache-2.0, so it is the one model here restricted to non-commercial use. Every use in this paper is non-commercial research, which all four terms permit. Supporting software: `transformers` and `PyTorch` (BSD-3-Clause), `scikit-learn` (BSD-3-Clause), `numpy` (BSD-3-Clause) and `matplotlib` (PSF-based license).

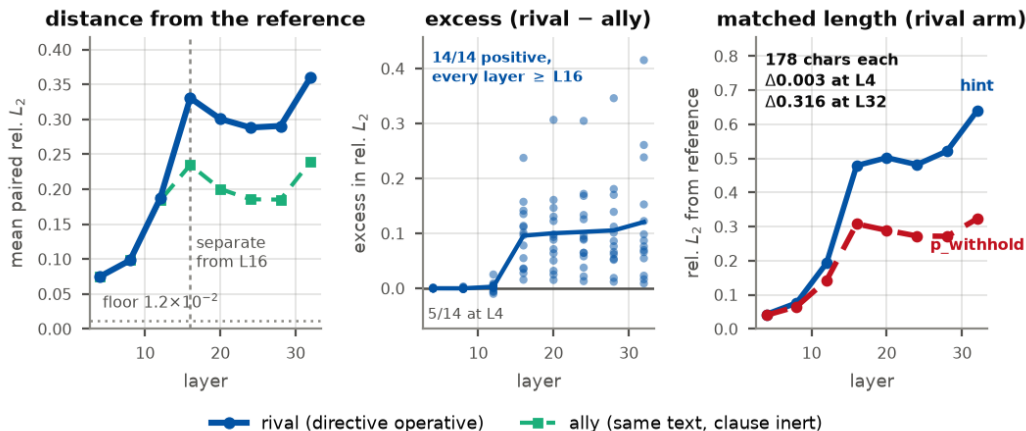


Figure 3: **The geometry is text shallow and role-dependent deep.** Left: mean paired relative L_2 from the reference variant across depth, rival arm (solid) against ally arm (dashed), with the imported numerical floor. The arms coincide through layer 12 and separate from layer 16. Centre: the per-variant excess (rival – ally); every one of the 14 variants is positive from layer 16 on, and the shallow layers are a clean null (5 of 14 positive at layer 4). Right: the length-matched pair *hint* versus *p_withhold*, identical at 178 characters, which sit 0.003 apart at layer 4 and 0.316 apart at layer 32, so the deep separation is not length. Llama-3.1-8B instructed, $n=400$ paired episodes per variant.

Two obligations attach to distributed *weights* rather than to experiments, and they are easy to miss: the Llama 3.1 license requires that a model fine-tuned from Llama Materials carry Llama at the start of its name and display “Built with Llama”, and both the Llama and Gemma terms require the accompanying notice and use policy to be passed downstream. We release code but no weights or adapters, so neither obligation binds this release; anyone redistributing an adapter trained with this code incurs both.

B The reward basin and its correction

This is the material §7 summarises.

Saturation on the wholly unmodified recipe: one family of four attempted. Decorrelating the emitted token from the semantic action makes the *role-conditional* policy substantially harder to learn, and our reward table gives role-insensitive policies well-separated payoffs: at a 50/50 role mix, always-truthful earns -0.50 , always-deceptive earns $+0.25$, and only the conditional policy earns $+1.25$. Three of four families never reached the conditional solution, and the three failed in *different* ways. Qwen-14B held ally truth 1.000 with rival deception ≈ 0 and reward pinned at exactly -0.50 for 340 epochs, then fell into the always-deceptive basin: rival deception 1.000 purchased by destroying the compliant branch (ally truth 0.000–0.500), so it is not a conditional deceiver and we did not probe it. Llama-8B and Mistral-7B landed somewhere we did not anticipate and initially mis-described: they keep a perfect compliant branch (1.000) but emit a single constant answer to rivals, which scores ≈ 0.5 deception purely because the secret bit is balanced (§3.3). That is a third basin, not the always-truthful one.

Correcting the payoff asymmetry. Replacing the table with a symmetric one ($+1$ for the role-appropriate answer, -2 otherwise, regardless of role, so always-deceptive earns -0.50 and is no longer positively rewarded) brings Qwen2.5-14B to deception 0.996 with ally truth 0.994, a conditional policy on both branches, with the identification result unchanged (§3.3). This has three practical consequences. First, an asymmetric adversarial payoff makes unconditional deception *positively* rewarded and therefore an attractor whenever the conditional solution is out of reach, a reward-design hazard worth flagging independently of probing. Second, anyone replicating our headline cell should expect the training half to be the hard part, and should verify **both** branches before probing: an arm at rival deception 1.000 with a broken compliant branch looks saturated on the headline metric while violating the premise that makes an ally-fit probe meaningful at all. Third,

Table 3: The aliasing rate under the instruction-pair protocol, measured on our task. γ is $P(c_{\text{dir}} = c_{\text{lie}})$ on adversarial trials; 1 means the two candidate targets are one label vector and no in-distribution check can prefer either. *Lying rates are shown to the precision that reproduces the γ column; the top rows are rounded to three decimals, so recomputing Eq. 2 from those digits can differ from the reported γ in the fourth decimal. The agreement asserted in code is between the measured γ and the closed form on unrounded rates, to within 10^{-9} .*

arm and instruction rung	$d_{\text{honest}} \rightarrow d_{\text{deceptive}}$	Eq. 2	measured
Gemma-9B, default rung	0.000 $\rightarrow$ 1.000	1.0000	1.0000
Qwen-14B, default rung	0.000 $\rightarrow$ 1.000	1.0000	1.0000
Gemma-9B, softened rung	0.000 $\rightarrow$ 0.890	0.9450	0.9451
Llama-8B, default rung	0.118 $\rightarrow$ 0.688	0.7850	0.7848
Qwen-14B, hint rung	0.000 $\rightarrow$ 0.553	0.7765	0.7764
Gemma-9B, hint rung	0.000 $\rightarrow$ 0.038	0.5190	0.5190
Gemma-9B, 5-wording bank	0.1211 $\rightarrow$ 0.8503	0.8646	0.8646
Gemma-9B, 5-wording bank, $N=1000$	0.1113 $\rightarrow$ 0.8669	0.8778	0.8778

and symmetrically, an arm near deception 0.5 may not be a partial deceiver at all: check the rate against the ground-truth bit before calling it one.

C The design rule and its feasibility floor

The instruction-pair protocol carries a feasibility floor as well as an aliasing rate (App. D). The identifying signal lives entirely in the minority side of whichever split is intended, so write p for the fraction of adversarial trials that side consumes. With a minimum viable subset size $n_{\min}$ and N_{adv} adversarial evaluation trials, the split needs $\min(p, 1 - p) \cdot N_{\text{adv}} \geq n_{\min}$, so for the usual regime $p > 0.5$,

$$N_{\text{adv}} \geq n_{\min} / (1 - p). \quad (1)$$

Which rate plays the role of p is fixed by the split, and mistaking one for the other is the easiest way to misapply the rule. For an *agreement* split over both matched rungs, $p = \gamma$: at $n_{\min} = 25$ the floor is 100 trials at $\gamma = 0.75$, 500 at 0.95, 2500 at 0.99, and unattainable at any N when $\gamma = 1$. For a *compliance* split, which can use only the disobedient trials of the deceptive rung, $p = d_{\text{deceptive}}$, and the pooled γ is the wrong quantity because it also counts honest-rung disagreements the split cannot touch. App. J reports a family where $\gamma = 0.985$ while $d_{\text{deceptive}} = 1.000$, so the floor is finite in γ and unattainable in the rate that actually governs. This is a screen, not a power calculation: the realized disagreement count is random, and clearing the floor guarantees neither class balance within the subset nor power for an AUROC contrast. Its value showed up early (a compliance split with no minimum size produced a striking number computed on ~ 8 trials), and it correctly predicted that our own $N=300$ split was infeasible. One refinement that matters: the governing rate must be measured on the *same subset* you intend to split, not pooled. Our pooled γ under-predicted the disobedient count in the held-out cell by 47%, because the held-out wordings drew more disobedience than average.

D The instruction-pair protocol

D.1 The aliasing rate, in closed form

The two candidate targets are c_{dir} (“the deceptive instruction is present”) and c_{lie} (“the model is lying”). Unlike our ally case these are not identical in general, so the relevant object is the rate at which they agree. In a matched two-rung design, where every episode is rendered once under each instruction so the halves are exactly equal, restricted to the adversarial subset where the deceptive instruction actually requests a lie:

$$\gamma(c_{\text{dir}}, c_{\text{lie}}) = \frac{1}{2} [(1 - d_{\text{honest}}) + d_{\text{deceptive}}], \quad (2)$$

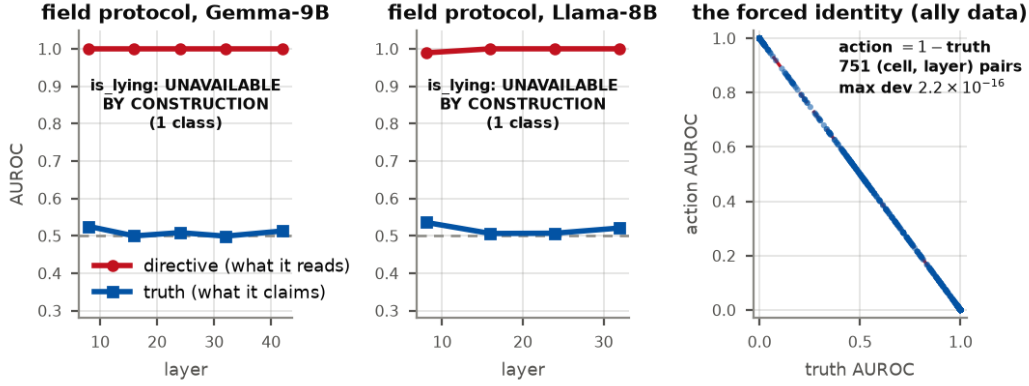


Figure 4: **The field’s instruction-pair protocol, run faithfully, beside the forced identity.** Left and centre: under the standard construction the probe reads the *directive* at ≈ 1.000 at every depth in both families, while *truth* sits at chance, and the identification split that would tell the two apart is unavailable *by construction*, because *is_lying* has only one class. Right: on ally data the action and truth AUROCs lie exactly on $\text{action} = 1 - \text{truth}$ across all 751 (cell, layer) pairs in our identification set, maximum deviation 2.2×10^{-16} , an algebraic consequence of the labels coinciding, not an empirical finding.

where d_r is the model’s lying rate under rung r on that subset. The proof is one line: under the honest rung the labels agree exactly when the model does not lie, rate $1 - d_{\text{honest}}$; under the deceptive rung they agree exactly when it does, rate $d_{\text{deceptive}}$; the matched design weights the halves equally.²

What Eq. 2 needs. No probe and no activations. Two behavioural rates, both of which a paper could report, determine how much of the probe’s target is the instruction rather than the lie. We release the audit as a 40-line function. Across seven rungs spanning three families our measured γ matches Eq. 2 to floating-point tolerance, and the code asserts the agreement instead of reporting it, because a mismatch would be a bug in one of the two computations and never noise. Run faithfully on two families, the construction reads the directive at ≈ 1.000 while truth stays at chance, and the identification split is unavailable *by construction* (Fig. 4).

D.2 Applied to published protocols

Table 3 gives the rates. $\gamma = 1.000$ is not a discovery about a probe; it is the statement “this model obeys both instructions”, and it would hold in a setup containing no probes at all. It is also the point at which this appendix and §3 coincide: the perfect aliasing that makes the conventional probe unidentified is this rate’s $\gamma = 1$ limit, so the two are one argument evaluated at two rates rather than two findings that happen to agree. The protocol is informative only strictly between the endpoints, and only if the rates are published.

Two of the four closest protocols, in their own words. These make the identification question unaskable by construction, and the two candidate readings of what their probe targets land at *opposite* extremes. Both force the response instead of sampling it: Goldowsky-Dill et al. [2025] insert “a random true fact. . . as the model’s response”, noting these “are not written by the model and contain no actual deception”, and Natarajan et al. [2026] state the response is “token-forced (off-policy)” with a classifier “trained to predict whether the honest or dishonest prompt was used”.

Under the *intent* reading, where the target is “this prompt asked for a lie”, forcing makes that label *definitionally identical* to the instruction, so $\gamma_{\text{intent}} = 1$ trivially and no in-distribution check can distinguish the two. Under the *content* reading, where the target is “what was said is false”, Natarajan et al. [2026] admits an exact result, because the *same* factual statement appears in both arms: response falsity is held fixed across the pair, so $d_{\text{honest}} = d_{\text{deceptive}} = f$ and $\gamma_{\text{content}} = \frac{1}{2}[(1-f) + f] = 0.500$ *exactly, for every f* : the instruction label is orthogonal to content falsity by design, as they say it is. Both endpoints are useless: at $\gamma = 1$ the instruction label *is* the lying label; at $\gamma = 0.5$ it is unrelated

²Three conditions, all met here and none automatic: the emitted answer is binary (no abstain state), the restriction is to adversarial trials only, and there are exactly two rungs. With unequal halves the general form is $\gamma = w_h(1 - d_{\text{honest}}) + w_d d_{\text{deceptive}}$.

to what was said. Neither leaves a target that is partly about lying. The two are easy to conflate: forcing does not make content falsity identical to the instruction, and quoting a bare “ $\gamma = 1$ ” for these protocols would assert exactly that.

The two rates, as published. Not one of the four publishes them. Two make the question moot by construction; one does not use the protocol; and the one design that could have reported them class-balances and discards them.

Three of the four state our mechanism qualitatively: Goldowsky-Dill et al. [2025] that the probe “partially depends on ‘is the scenario deception related’ ... instead of only on ‘is the response itself actually deceptive’”; Natarajan et al. [2026] that “the fundamental direction being distinguished may not be directly related to strategic deception”. Our delta is not the observation. It is the closed form, the diagnostic, and the design that breaks the aliasing.

D.3 Running the field’s construction on our task

We implement the protocol on our own game, including a faithful variant that closes all three deviations we would otherwise have to disclose: response-token mean-pooling excluding the last five tokens, token-forced responses, and a bank of five matched instruction wordings. We do not claim to reproduce any published result: we use none of their models, datasets, instruction banks or evaluation sets. The claim is about what the *protocol* inherits.

The identification test under the faithful construction. It is unavailable at every N , on both families we ran. Because the inserted response names no value, it misstates nothing: c_{lie} is the constant zero, so there is no lying anywhere in the sample for the instruction to be aliased *with*, the c_{lie} AUROC is undefined, and no sample size, model or arm makes the split available. Meanwhile the probe reads c_{dir} at 1.000. The protocol reports success by its own criterion while carrying provably zero information about actual lying. We flag one trap for anyone repeating this: the measured γ is exactly 0.500 here, and 0.500 elsewhere in this paper means “decorrelated, probe identified”. Under forcing it means the opposite kind of thing: it is simply what a constant scores, since half the adversarial rows carry $c_{\text{dir}} = 0$ and match a constant zero for free. It must never be quoted without the number of c_{lie} classes beside it.

Aliasing, high but wording-dependent. With one canonical instruction pair and an obedient arm, $\gamma = 1.000$. Averaged over five matched wordings on the same model and task it is **0.865 (0.878 at $N=1000$)**: the two candidate label vectors are $\sim 87\%$ identical, not 100%, so no single number should be quoted. Here Eq. 2 predicts a non-trivial value exactly.

E Related work, case by case

This is the material §6 summarises.

The umbrella claim, which we do not contest. That reward-trained deception can degrade passive probes is established [Taufeeque et al., 2026, Gupta and Jenner, 2025], as is probe fragility as a training-distribution artifact [Kumar, 2026] and the dominance of prompt choice in probe performance [Natarajan et al., 2026]. Sub-chance probe AUROC has been reported independently by two groups (0.376 and 0.374); the spread behind the first is stated in the introduction. Our contribution is the diagnosis, not the sighting.

Three results that have looked contradictory. All three are consistent with the identification account, a claim about their *form*, not a reproduction; we have not re-run these setups (§7). Goldowsky-Dill et al. [2025] report instruction-pair probes transferring at AUROC > 0.96 , long read as contradicting the inversion literature. Under our account it is a confirming instance: an ally-fit probe becomes indistinguishable from an action probe only once the policy reliably makes $\text{action} = \neg \text{truth}$, and their liars were not saturated. Our within-model pair brackets that relation, one arm at $\text{action} = \neg \text{truth}$ throughout and one with action independent of truth (§3.3), but neither arm is an unsaturated *conditional* liar of the kind they study, so the pair supports the argument’s endpoints rather than reproducing their regime. Taufeeque et al. [2026] report off-domain AUC $0.991 \pm 0.009 \rightarrow 0.376 \pm 0.261$ after RLVR with *no* detector penalty. Their account is representation drift, their term, alongside a behavioural measurement: the rate at which the model endorses hardcoding as acceptable rises from 10–40% to 45–65%, which they read as the model genuinely no longer considering hardcoding

deceptive. That is weaker than the claim that the truth ceases to be *represented*, which they do not assert. In our setting that stronger form is ruled out by the checkpoint-wise suppression result (App. G); and a below-chance number does not *require* either mechanism, because their probe was fit where truth and action were collinear too. We claim an alternative explanation is available, not that theirs is wrong in their setting. Natarajan et al. [2026] report both fingerprints we diagnose (a below-chance 0.374, and prompt choice explaining 70.6% of variance) and frame the first as performance headroom. Their positive conclusion is that instruction-pair probes capture intent rather than content; under §3 that is precisely the solution an unidentified ally-fit probe is expected to converge to, so their interpretation and their sub-chance AUROC are the same fact stated twice. Kirch et al. [2026] independently find the largest generalisation failures for behaviours defined by response intent.

On probing at the answer position. Natarajan et al. [2026] deliberately exclude the last few response tokens when pooling activations, to stop the probe reading truth off the output. Our probe sits *at* the answer-prediction position, which invites the objection that our inversion is output-token leakage the field already knows to avoid. Three responses. (i) $\text{truth/mixed} = 1.000$ on those same activations at that same position, so the position plainly carries the truth. (ii) We decorrelate the token by construction rather than by truncation, and fit token as its own label set: the fitted truth and token directions are near-orthogonal in every cell ($\cos = +0.003$ to -0.046 ; §2). (iii) Dropping those tokens removes $\text{truth} \leftrightarrow \text{token}$ while leaving $\text{truth} \leftrightarrow \text{intended action}$ untouched, and intent is upstream of them, so it is exactly the leg that survives. Probing at the output-adjacent position is also the point: it is where deployed monitors read and where the inversion literature reports its numbers.

Adjacent and converging. Adarsh et al. [2026] characterize context as applying a rotation-plus-scaling to the truth direction rather than destroying it, a geometric statement of why a probe fit under the wrong context distribution fails while the information remains present. Panfilov et al. [2025] flag instructed, off-policy probe evaluation as the untested case; Nyoma [2026] and Berger [2026] document adjacent failure modes; and Marks and Tegmark [2024] established the linear truth-direction structure these methods rest on.

F The apparent finding, and the control that dissolves it

This is the matched-control result §7 summarises.

On the Llama-3.1-8B anchor, a fixed-sign truth probe at the answer-prediction position collapses to AUROC 0.000 on the reward-trained arm while the instructed control holds at 0.985–0.998 across three seeds. The effect was seed-robust, localized to that position (intact at -3 , inverted at -1), and *causally* confirmed: patching the ally-context residual into a rival-context forward pass at mid-stack flipped the trained liar’s answer on 0.99–1.00 of trials versus 0.575–0.61 for the instructed arm. Read at face value this is a mechanistic difference between deception that was *trained* and deception that was *told*, on exactly the axis safety cares about.

It is not. The instructed arm lies on only 0.735 of rival trials and answers correctly on just 0.902 of *ally* trials: a worse deceiver and a worse player. Build behaviourally matched pairs (both arms at ally truth 1.000 and rival deception 1.000, available in Gemma-2-9B and Qwen2.5-14B) and the difference vanishes on five of six measures: the answer-position readout (both 0.000, an exact tie in all 8 matched cells); patch magnitude (1.000/0.955 and 0.960/0.960); patch transition depth and shape (same window, same sharpness); ablation of the truth direction (1.000/1.000, null in both); and a steering α -sweep at 5/10/20/40 (0.000/0.000, null in both). The sixth does not vanish, and it runs against the hypothesis. On the off-position (-3) readout the matched difference is consistently *positive* (8 of 8 cells, mean $+0.12$, 95% CI $[+0.055, +0.190]$), meaning the *reward-trained* arm is the *less* inverted one away from the answer slot, the opposite sign to an account in which training hides the truth. Its absolute level is also wildly family-dependent (both arms sit near 0.00 in Gemma-9B and near 0.90 in Qwen-14B/32B), so it is not a stable quantity to compare across families in the first place.

The unmatched instructed arm’s roughly-chance patch response and non-saturating ramp (asymptoting at ≈ 0.72) are confounded with its lower behavioural reliability rather than evidence of a distributed representation; we did not measure an intervention-specific ceiling, so we rest the argument on the matched arms, which simply do not reproduce the difference. And at the one layer where the matched Gemma arms differ at all the *instructed* arm is ahead (0.355 vs 0.100 at L25), the opposite sign to

the hypothesis. Across a 7-family $\times$ 2-arm $\times$ 3-seed grid at $N=1000$, all 21 emergent grid cells are identical to four decimals (deception 1.000, AUROC 0.000, cross-seed sd 0), and instructed arms invert too, sometimes *more* (Gemma-9B instructed 0.015 versus its emergent counterpart 0.165). We state this as a null with tight bounds, not proven equivalence (Fig. 7). So what *does* predict inversion? We tried lie rate, then per-example confidence, and refuted both (App. G). The question was ill-posed: the quantity being predicted was never a property of the model’s representation.

G The honest negatives

These are the honest negatives §7 summarises.

We were wrong twice, and the negatives constrain §3.

Neither lie rate nor per-example confidence: our own successive hypotheses (Fig. 8). Mistral-7B instructed reaches rival deception of exactly 1.000 and does **not** invert (AUROC 0.970), while other arms at 1.000 sit at 0.000; a rate-based account predicts the opposite. Mistral also has *lower* rival entropy (0.252) than the Llama-8B variant that does invert (0.283), so it is the more deterministic of the two and the one that stays at 0.970. Within a single model the ordering fails too: entropy falls monotonically across a directive ladder (0.490, 0.468, 0.324, 0.305, 0.283) while AUROC goes 1.000, 0.790, 0.998, 1.000, 0.080. Entropy does not order the data, and the ladder confounds directive *text* with resulting behaviour, so it could not have settled the question either way.

Suppression, not loss of the representation. Along a full RL trajectory (41 checkpoints for seed 0 and 36 each for seeds 1 and 2, the latter two evicted before the flat post-collapse tail), mid-stack decodability stays pinned at 0.86–1.000 at *every* observed checkpoint, including every point at which the final layer has collapsed to 0.000; in seed 0 it reads 1.000 at all 41 (Fig. 9). Training never removes the linear signal mid-stack; only the answer-prediction readout flips. This rules out the strong form of what we will call *belief drift*, that the information stops being represented, which is the form that would explain a below-chance probe. The strong form is our own formulation, not a claim we attribute to others: the nearest account in the literature [Taufeeque et al., 2026] claims representation drift together with a measured shift in how the model evaluates the behaviour, *not* the absence of a linear representation. Our result does not adjudicate weaker versions in which the information is retained but no longer operative.

A cliff, with seed-dependent onset and width. AUROC is pinned at 1.000 across measured deception from 0.03 up to at least 0.90 in all three seeds: a model that lies to rivals on 53% of trials reads 1.000, perfectly truthful. But it is not a fixed threshold and not uniformly abrupt: seed 0 begins degrading at 0.938 and passes through 0.959 $\rightarrow$ 0.828 and 0.964 $\rightarrow$ 0.372 before reaching 0.000, roughly six percentage points, whereas seeds 1 and 2 hold 1.000 through 0.981 and 0.984 and collapse within one to two. We report per-seed onsets (0.938, 0.986, 0.990) instead of a pooled curve, and treat the variation as further evidence that aggregate rate is the wrong independent variable. Out of distribution on an unseen game structure the *same adapter* loses its inversion (0.000 $\rightarrow$ 0.550) exactly when its own deception falls (1.000 $\rightarrow$ 0.559), though the out-of-distribution points do not lie on the in-distribution curve, so the readout is **not** a pure function of deception rate across tasks and we do not draw it as one (App. A).

Report spread, not point estimates. The readouts carrying the App. F null sit at extremes where the measurement is exact: all 21 saturated emergent grid cells give cross-seed sd 0, the four inverted cells 50-resample sd 0.0000. The interpretive middle is softer: our worst cell reads 0.205/0.479/0.531 across three episode seeds with a 50-resample bootstrap of **0.539 $\pm$ 0.070**, approximately chance, and *not* the inverted cell any one of those point estimates would suggest. This is structural: at AUROC 0.000 or 1.000 the classes are perfectly separated so any subsample reproduces the ordering, whereas intermediate values carry genuine sampling variance. Intermediate cells should be reported with spread or a CI, by us and by others.

H The constructive result: further detail

This is the material §3.3 summarises.

Per-seed collapse profiles for row 1. In the representative seed the ally-fit probe begins collapsing at layer 26 (0.354), is at 0.008 by layer 27, and sits at 0.000 from layer 29 through layer 42 of 42;

the other two seeds reproduce the profile, intact through layer 16 and degrading at layer 24 (0.866, 0.853).

What the second arm actually is. Reading the deception rate alone invites the description “the same run, less far along”. It is not. Breaking that arm’s rival behaviour down *by the true bit*, which we did only when a later causal run printed the split, shows it answers truthfully on **0.000** of true-bit-0 rival trials and **1.000** of true-bit-1 ones: it emits a constant semantic answer to rivals, while still answering allies truthfully at 1.000. Its “deception rate” is therefore $P(\text{truth} \neq \text{that constant}) \approx 0.5$ *by arithmetic*, on any draw. The Mistral-7B row (row 7) is the mirror image, constant at the other value.

Why a constant branch sharpens the contrast. Our account says inversion requires $\text{action} = \neg \text{truth}$ on the evaluation trials, and a constant rival branch is the *maximal* break of that relation, since action is independent of truth, so an ally-fit probe should read truth well there, which is what 0.815 and 0.969 are. The two arms bracket the action–truth relation: row 6 is not a less reliable liar, but a policy whose rival action is independent of the truth. *One consequence:* this pair therefore does not illustrate the unsaturated-conditional-liar regime that Goldowsky-Dill et al. [2025] occupy, so our reconciliation with them (§6) rests on the argument and on the instructed-arm ladder, not on this pair.

What the fixed-symbol guard misses. We verify the policy is not degenerate with the fixed-symbol statistic of §2, and both arms pass it (0.516 and 0.484, both ≈ 0.5), for exactly the reason flagged when we introduced the statistic: it rules out a fixed surface character and is blind to a fixed meaning. The cheap fix is to report the deception rate split by the true bit: a conditional deceiver has both components away from the extremes, a constant-output policy reads 0.000/1.000. An aggregate rate near 0.5 is exactly what a degenerate policy produces, and it reads as healthy graded behaviour.

Row 7 also carries a result independent of all this: its token/mixed climbs to 0.975 at the final layer while truth/mixed stays at 1.000, which shows that truth and the surface symbol coexist *separably* in the final residual stream, against the reading that late layers are “just the output token”.

Why only one family reached the full conjunction. Row 1, the only cell combining reward training, deception 1.000, training on the identification task *and* a wholly unmodified recipe, holds in one family, across three seeds. That precise conjunction is still one family and we do not claim otherwise.

But the reason the others failed is not probe-related, and it was actionable: our reward table makes unconditional deception an attractor whenever the conditional solution is out of reach, which is why three of four families never learned one, and correcting that single design flaw rescues one of them (§7 gives the table and the arithmetic). Under the corrected table Qwen2.5-14B reaches rival deception 0.996 with ally truth 0.994, a genuinely role-conditional policy on both branches, and shows exactly the pattern of the headline cell: truth/ally collapses from 1.000 at layer 24 to **0.000** at layer 40 and **0.023** at layer 48 of 48, while truth/mixed reads **1.000** at every layer measured.

Why this is more than an extra row. First, the same corrected table run on Gemma-9B reproduces its original-table result exactly (truth/ally = 0.000, truth/mixed = 1.000), so the identification phenomenon is invariant to the reward table: the change alters what is *learnable*, not what the probe reads, and the new cell is therefore comparable to the others. Second, the rescue was derived from our own basin analysis rather than found by search, so the limitation comes with a mechanism *and* a demonstrated fix. The saturated set now spans three architectures (Gemma-2-9B, Llama-3.1-8B, Qwen2.5-14B) and four recipes, and the deepest of them is a 48-layer stack. The constructive result never depended on this anyway: truth/mixed = 1.000 holds in 19 of the 21 identification cells including the instructed arms.

I The inferred-truth task, settling and ingredient controls

This is the material §7 summarises.

Partial evidence, at one cell, that the direction is not reading a copied token. We ran one probe against this limitation. In an *inferred-truth* variant of the codebook task the secret bit is never a token in the prompt: it is the outcome of a stated comparison between two quantities (operands separated by at least 31, never 0 or 1, both rule orientations, so the bit cannot be read off operand magnitude alone). On a Gemma-2-9B instructed arm at rival deception 0.545, the mixed-fit probe reads truth AUROC **1.000** at layers 24, 32, 40 and 42 of 42. The depth profile is itself the evidence (Fig. 12):

a bit the prompt *states* is readable by layer 4, whereas this one sits at chance through layer 16 and only then rises; a copied token would not behave that way. Alongside this the fixed-symbol guard sits at ≈ 0.5 ; the ally-fit probe never inverts (0.374–0.551), as an unsaturated policy should not, and the forced identity holds at all seven sampled layers. The probe recovers the bit at 1.000 while the arm itself answers correctly on only 0.733 of compliant trials, so the bit is linearly present even on trials the model gets wrong. We report this as *bounding* rather than settling the limitation, and it is two removes from a headline: the arm is instructed, and its compliant accuracy is 0.733. A third concern, that a linear map over activations encoding both operands could in principle perform the comparison *probe-side*, is the substantive one, and we take it up below: for the mixed-orientation fit the task’s own construction makes that map unavailable, and we measure that its ingredients are linearly present while the bit is not. So this cell rules out a copied token, and whether the policy computed the bit is addressed next. **The settling experiment, which localises the confound rather than excluding it.** We fit the mixed probe on two of the four inferred-truth templates, froze it, and scored it on the two held-out templates (different domains, variable names and comparison wordings) with per-trial codebook randomisation decorrelating the emitted symbol, and with a same-task control arm frozen from the same probe and scored on the fit templates at a fresh seed ($N=2000$ per arm). The three subsets are matched on capability (compliant accuracy 0.764 fit, 0.741 held-out, 0.785 control), so a drop cannot be attributed to the model failing to compute the bit. Over the layers where the control carries signal at all, the frozen probe peaks at $\text{truth/mixed} = 0.984$ at layer 28 against a control of 1.000, a transfer cost of -0.016 , and holds 0.962/0.942/0.931 at layers 32 / 40 / 42 ($-0.038 / -0.058 / -0.069$). Freezing is not free, and the cost is strongly depth-dependent: it is -0.285 at layer 20 and -0.221 at layer 24. So a frozen probe is measurably worse than a refit one on a new task, by an amount that depends on where it is read (Fig. 5).

Why this task forecloses the probe-side alternative. That concern turns on what the label actually is. Because exactly one operand is drawn from a high band and one from a disjoint low band, the truth bit is not a general magnitude comparison: writing S for “the first slot holds the high-band value”, the bit is S under the “greater” rule and $\neg S$ under “lesser”. The label is therefore the exclusive-or of S with the rule orientation, exactly and by construction. An exclusive-or of two features is not linearly separable in those features, so no single direction combines a representation of S with a representation of the rule word into the bit. A linear probe cannot perform this comparison from the raw ingredients; something must already have combined them, and at the final prompt position combining information carried by the operand and rule tokens requires attention, that is, the policy’s own computation. The one residual case, that the activations happen to carry a nonlinear interaction of the two factors which a linear probe can then read off, is not the confound but an instance of the claim.

The two ingredients, measured. The argument above would be empty if S and the rule orientation were not linearly present at the read position to begin with, so we probe for each of them separately: same cell, same nine layers, same $N=2000$ and the same seed as the fit arm above. Both read AUROC 1.000 at layers 8 and 16, exactly where the bit itself is at chance (0.442 / 0.530), and 0.934 / 1.000 already at layer 4. So the information a probe-side comparison would need is *perfectly* linearly available, and a linear probe still cannot produce its exclusive-or: the alternative is demonstrably unavailable to a linear map on this task rather than merely unproven. The single-orientation arm doubles as an end-to-end check on the S label, since with the rule pinned the bit reduces to S exactly and the two AUROCs must coincide: they agree to 0.0×10^0 at all nine layers. The rule’s decodability is close to lexical, the orientation being a phrase in the prompt, so its 1.000 is unsurprising and is **not** a finding about the model; its only role is to establish that the ingredient is there for a probe to use. And the scope does not widen: one model, one arm, one task family.

The single-orientation arm as positive control. Fixing the rule removes the exclusive-or: with “greater” pinned, the bit simply *is* S , a shallow feature available from operand magnitudes without reading the rule. A probe fit there reads 1.000 at layer 8, where the mixed-orientation fit sits at chance (0.442 / 0.530 at layers 8 / 16), and when frozen and scored on flipped-rule trials it *inverts*, reaching exactly 0.000 at layers 16 and 20 and staying below chance from layer 4 to layer 24, against a control at 1.000. So the instrument detects probe-side computation when it is possible, and reports its absence when it is not. Both arms then recover at layer 28 (0.939 contaminated, 0.984 clean) before easing to ≈ 0.82 and ≈ 0.93 at the final layer. On the clean fit a probe *refit* on the held-out templates is also at chance shallow (0.474 / 0.496), so no shallow signal exists for any probe to read; on the contaminated arm a refit probe reads 1.000 from layer 16 up, so what fails there is the frozen

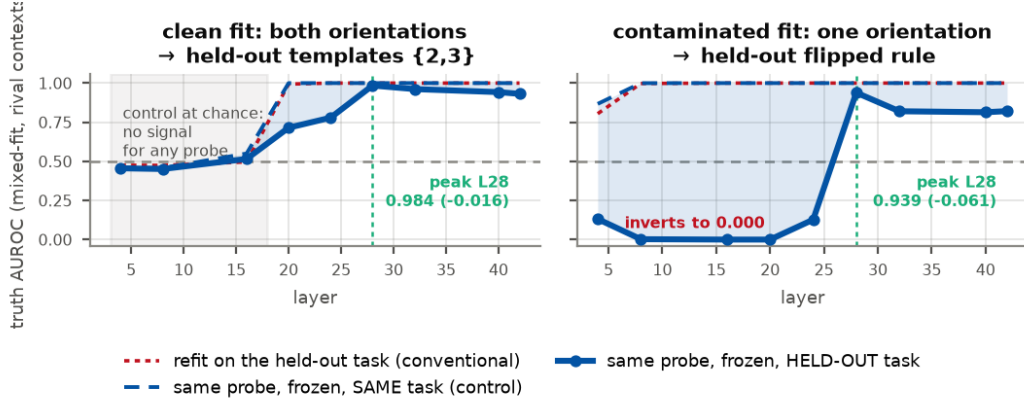


Figure 5: **The settling experiment.** The same frozen probe scored on a held-out task (solid) and on its own same-task control (dashed); the shaded gap is the transfer cost, and the dotted line is what refitting on the held-out task would give. Left, clean fit: with both rule orientations in the fit the bit is the exclusive-or of two prompt facts, so no shallow shortcut exists: the frozen probe is at chance shallow, where a *refit* probe is also at chance (0.474/0.496), and transfers at **0.984** (−0.016) at layer 28, 0.962/0.942/0.931 at layers 32/40/42. The cost is depth-dependent and worst at layer 20 (−0.285; layer 24, −0.221). Right, contaminated fit: restricting the fit to one orientation removes the exclusive-or, the bit becomes the shallow feature “which slot is larger”, and the frozen probe *inverts* to 0.000 at layers 16 and 20 under a rule flip: the probe-side confound, measured. Both arms peak at layer 28 and ease off. Grey spans mark layers where the control itself is at chance, so a small gap there reflects no signal rather than good transfer. Gemma-2-9B instructed, $N=2000$ per arm, capability matched (0.764/0.741/0.785).

direction, not the held-out task. An inversion is a stronger signal than chance, so neither shallow result can be an artifact of weak capability. This supersedes rather than fulfils the prompt-only baseline our own specification asked for: the confound that baseline was meant to detect is not linearly available to begin with, and the single-orientation arm exhibits what its signature looks like when it is.

Two by-products of building that task. First, a frozen instruction-following limit: base models compute such a comparison reliably under the plain rule but not under an *inverted* one, in all three families we tried (App. A). They answer from operand magnitude and largely ignore the comparison word, which is also why the cell above is capped at compliant accuracy 0.733. Second, a learnability finding: RL did not learn a role-conditional policy over an *inferred* bit in any of three attempts, across two families and both task variants, with no convergence to any reward basin, which is why the cell above is an instructed arm. We frame this strictly as a fact about what this reward signal makes learnable in this game, **not** as evidence about truth representations.

J Wording generality and compliance tracking

These are the two objections to the instruction-pair readout of App. D.

Is the instruction readout merely a two-string discriminator? With a single fixed pair, “which instruction was present” is decidable from any surface difference between two fixed strings. We settle this with leave-one-wording-out over a five-wording bank: five fits, each trained on four wordings and scored on the fifth, under a doubly-held-out split that partitions *both* episodes and wordings and uses only the diagonal cell. Every wording is held out exactly once, so the spread across folds is a wording-generalization spread, not a resampling artifact.

wording held out	L8	L16	L24	L32	L42 (final)
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
2	0.969	1.000	1.000	1.000	1.000
3	1.000	0.825	1.000	0.998	0.942
4	0.404	0.501	0.991	0.974	0.867
range	0.596	0.499	0.009	0.026	0.133

907 **A depth-dependent answer, against a pre-registered threshold** (a range of 0.05 or less counts as
 908 wording-invariant). *At layers 24 and 32 the readout is wording-invariant:* range 0.009 and 0.026,
 909 minimum 0.974, so every wording is classified at least that well by a probe that never saw it. The
 910 two-string objection is answered there. *In the shallow stack it is not:* one wording sits at 0.404
 911 and 0.501, i.e. at or below chance. Below chance is systematic inversion rather than mere absence
 912 of signal: AUROC is antisymmetric under label negation, so a reversed-sign readout of the same
 913 direction would score 0.596. (AUROC is a ranking statistic, so this is a statement about ordering, not
 914 a classification rate.) So “the instruction signal generalizes across surface form” holds from layer 24
 915 up and is a per-wording accident before that. The final layer is intermediate (range 0.133, minimum
 916 0.867) and we report it as a range, not a point.

917 **Why the shallow readout fails.** Four of our five wordings distinguish their honest from their
 918 deceptive half with a *lexical polarity* cue (truth/falsehood, correct/wrong, disclose/withhold); the fifth
 919 was deliberately built with no negation vocabulary on either side, separable only by the value mapping
 920 itself. That suggests the shallow probe latches onto the cue, and it makes a falsifiable prediction about
 921 a cross-wording transfer matrix, which we pre-registered and then ran: fit on wording *i* alone, score
 922 on wording *j* alone, all 20 ordered pairs, episodes held out throughout. *At layer 8 the prediction holds*
 923 *on all three clauses:* transfer among the four cue-carrying wordings is uniformly high (all ≥ 0.801 ,
 924 mean 0.948), while the cue-free wording is unreadable from any of them (column mean **0.497**) and
 925 offers nothing in return (row mean **0.463**). The most extreme cell, fitting on the cue-free wording
 926 and scoring on the strongest-cue one, reads **0.053**, a strong rank inversion, the signature of a cue that
 927 is present but points the wrong way.

928 Above layer 8 the asymmetry *reverses*, and we flag this as a post-hoc observation rather than a
 929 registered result: by layer 16 a probe fit on the cue-free wording reads the others at 0.928–1.000
 930 while they still cannot read it (column mean 0.596), and by layer 24 the matrix has largely converged
 931 (worst cell 0.699). The cue-free wording is not therefore the best one to train on: by mean transfer
 932 it is the *worst* of the five at three of four depths. The defensible claim is narrower: it is the most
 933 diagnostic wording, and the five folds must not be averaged as though exchangeable, since only one
 934 of them tests generalization to a cue-free target.

935 **The trough on a denser layer grid.** It is at layer 12, not layer 8, and there the cue-free wording
 936 is inverted in *both* directions (row mean 0.233, with individual cells as low as 0.092; column mean
 937 0.397). Recovery is then abrupt: by layer 20 the matrix is fully converged (every cell 1.000, column
 938 mean 0.976) before drifting down again in the last third of the stack.

939 **The inversion is not an evaluation-sampling artifact.** With episode-level bootstrap intervals on
 940 every cell (2000 resamples), all 20 cells at each of layers 8, 12 and 20 have 95% intervals that exclude
 941 0.5, including all eight inverted cells at layer 12, which are tight and far from chance: the cue-free
 942 row reads [0.12, 0.14], [0.08, 0.11], [0.30, 0.32], [0.39, 0.40] and its column [0.39, 0.41], [0.45, 0.46],
 943 [0.43, 0.44], [0.28, 0.31]. *So the trough is a real feature of this fit rather than noise.* The bootstrap
 944 resamples held-out episodes while holding the fitted probe, the training wordings and the partition
 945 fixed, so it measures evaluation variability at a fixed fit and is *not* a statement about refit stability. The
 946 two are distinct quantities.

947 **Two further families: the ordering replicates, the inversion does not.** Running the same matrix
 948 on Llama-3.1-8B, the cue-free wording is again the hardest target at every depth (column mean
 949 0.53–0.75 against a cue-carrying block of 0.72–0.999), but no cell anywhere falls below 0.500.
 950 Qwen2.5-14B ($N=2000$, seven depths) gives the same verdict: the cue-free column mean is the
 951 lowest of the three at *every* depth (0.507, 0.540, 0.570, 0.620, 0.861, 0.806 at layers 4–48, against a
 952 cue-carrying block reaching 0.99), while the cue-free *row* mean never drops below 0.566 and nothing
 953 inverts from layer 8 up. So “four of five wordings share a polarity cue and the cue-free one is the
 954 hardest target” holds in all three families we tested, while “the shallow probe latches onto the cue
 955 hard enough to invert” holds in one of three and is, on present evidence, specific to Gemma, as is the
 956 *location* of the anomaly, since Qwen’s worst depth is its shallowest rather than a mid-stack trough.
 957 *One caveat against over-reading Qwen’s shallowest layer:* at layer 4 its matrix has eight sub-chance
 958 cells scattered across the cue-carrying block itself (as low as 0.081) and its grouped directive readout
 959 is only 0.824, so that layer is generally unstable rather than specifically cue-driven. The negative
 960 rests on layers 8 and above, where the matrix is well behaved.

961 **Which layers this claim may be anchored at, measured two ways.** Refitting the probe under five
 962 different split seeds (a new held-out wording set and a new episode partition each time) gives an

across-refit sd of 0.0045 at layer 8, **0.081** at layer 16, **0.0099** at layer 24, 0.028 at layer 32 and 0.035 at layer 42. The leave-one-wording-out folds above rank the layers identically (range 0.009 at layer 24 and 0.026 at layer 32, against 0.50–0.60 shallow), so two independent instruments agree that layers 24–32 are where this readout is stable and layers 16 and 42 are where it is not. This also settles an apparent contradiction in our own numbers. The final-layer value moved from 0.965 to 0.849 to 0.897 across three runs, a range of 0.116, while its *bootstrap* sd was 0.0076, apparently a twelve-fold contradiction. The refit sd at that layer is 0.035, with a range of 0.102: the run-to-run movement is ordinary refit variance, and the bootstrap simply could not see it, because it resamples evaluation episodes while holding the fitted probe and the partition fixed. An interval that omits the dominant variance component is not a stability claim, and we quote the refit spread instead.

Two concessions. The ungrouped 1.000 was measured on a split in which 100% of test rows had their matched twin in training: with a matched-pair design the row-level split is maximally leaky, because the twin differs only in the label. And this is five wordings from one bank on one model and one task; the invariance we demonstrate is over those five. The matrix carries per-cell intervals but comes from a single fit, so it is evidence consistent with the lexical-cue mechanism rather than a uniquely identifying test of it: a different set of cue-carrying wordings could produce the same pattern for a different reason.

Does the instruction-fit direction track who actually lied? This is the question the protocol implicitly claims, and on a non-leaky split our answer is no at the layers that matter. We restrict to the adversarial subset and split on compliance. Expectation is not what makes that split available: $\gamma < 1$ leaves a disobedient fraction on average, but the realized counts are what decide it, and the held-out cell contains 238 complied and 52 disobeyed trials, clearing our minimum of 25 in both halves. Compliance AUROC, with episode-level bootstrap intervals over 145 episodes:

layer	8	16	24	32	42 (final)
held-out split	0.705	0.640	0.689	0.593	0.365
95% CI	[0.613,0.793]	[0.568,0.707]	[0.632,0.741]	[0.534,0.653]	[0.293, 0.438]
row-level (leaky)	0.721	0.813	0.840	0.806	0.736

The leak inflated this quantity by up to 0.37, the largest leak-induced distortion we measured, for the reason given above. But the decisive observation is not the leak.

Non-identifiability across instruction wordings: the sign reverses. Repeating the compliance split inside each leave-one-wording-out fold, it is available in only two of the five, and those two disagree in *direction* at four of five layers:

wording held out	L8	L16	L24	L32	L42 (final)
3	0.325	0.477	0.781	0.652	0.607
4	0.638	0.483	0.394	0.200	0.080

At the final layer one fold scores compliance at AUROC 0.607 [0.542, 0.670] while the other scores it at 0.080 [0.048, 0.113], which by antisymmetry is an AUROC of 0.920 for *disobedience*. The intervals are tight and nowhere near overlapping. We therefore make no claim that the instruction-fit direction tracks who actually lied, in either direction, at any layer. A quantity whose sign depends on which instruction wordings the probe was fitted on is not a measurement of the model.

The paper’s own diagnosis, inside the field’s protocol. Where §4 has two probes on the same activations returning opposite verdicts under different prompt variants, here refitting on a different subset of *instruction wordings* reverses a compliance prediction. We state it as differently-fitted directions disagreeing: the folds train on different wording sets, so these are different fits, not one unstable direction. We offer no mechanism for it.

The availability pattern as the design rule in action. The split needs disobedient trials, and the per-wording disobedience rate spans **0.000** to **0.309**: two of the five wordings draw *perfect* compliance, so their identification test would need ~ 2550 adversarial trials rather than the ~ 81 the most disobedient wording needs. *The pooled rate*, 0.128, *predicts none of the five*. On a second family the whole protocol falls off the same cliff: Qwen2.5-14B complied with the deceptive instruction on 1490 of 1490 adversarial trials, pooled over all five wordings, so $d_{\text{deceptive}} = 1.000$ and Eq. 1 returns *unattainable at any N*: the compliance split is unavailable on that family for a pure design reason,

with no probe fitted. Its aliasing rate is nonetheless non-degenerate ($\gamma = 0.985$ with $d_{\text{honest}} = 0.029$, so Eq. 2 is doing real work rather than reducing to an endpoint), which is precisely the combination the rule is meant to catch: high but sub-unit aliasing that still leaves no testable subset. Anyone applying Eq. 1 must evaluate it on the subset they intend to split, and against the fraction that split actually consumes: for a compliance split that is $1 - d_{\text{deceptive}}$, not the pooled $1 - \gamma$, which also counts honest-rung disagreements the split cannot use.

K The causal analysis

This is the causal analysis §7 summarises.

Decodability, not causal use. truth/mixed = 1.000 says a linear map recovers the bit, not that the model uses it. We tested both directions causally, the conventional ally-fit one and the mixed-fit one this paper hands back, at the layer where the ally-fit readout is fully collapsed, on the task the direction was fitted on, at $N=1000$ per condition.

Steering is null, and on two arms. Sweeping $\alpha = 5/10/20/40$ moves nothing. On the saturated reward-trained arm the rival-truth rate is 0.000 at every α , which bounds the effect at 0.3 percentage points per tested condition (exact one-sided 95% bound on 0/1000). *Per condition* is the qualifier that matters: eight conditions were inspected (4 values of $\alpha \times$ two sign modes), so this is not a simultaneous 95% bound over the sweep, and the conditions share episodes and are therefore not independent tests to be pooled. We also note this arm sits on a floor, since its unperturbed rival-truth rate is already 0.000, so only an increase was detectable. The instructed arm has no such floor (its baseline is ≈ 0.20) and there the rate is flat at 0.214–0.226 across an eightfold range of α (0.65 sd; a genuine linear steer would scale with α). We also ran a signed variant that takes its sign from the ground-truth label, an oracle no deployed detector could have, precisely because unsigned steering could cancel across trials. It is identical to the unsigned arm at every α , to all reported digits, and that agreement is mostly forced rather than measured: on true-bit-1 trials the oracle sign is +1, so the two arms are literally the same intervention there, and on true-bit-0 trials, the only trials where they differ, the rate is pinned at 0.000 in every condition of both arms. So the signed arm does not independently corroborate the null. What it does establish is narrower: on true-bit-0 trials $-\alpha d$ fails to move the rate off its floor just as $+\alpha d$ does, so on that subpopulation both directions were tested and both are null. The cancellation explanation is excluded only there, and the decisive test, $-\alpha d$ on true-bit-1 trials, we have now run, along with the ally-fit comparison; both are reported in §K.2 immediately below, and the depth sweep that was previously missing is in §K.3.

K.1 The two directions separate exactly where the collinearity breaks

Before the causal comparison, a geometric one that costs nothing. Our probe files store the ally-fit and mixed-fit coefficient vectors for the same target at the same layers, fitted by the same code on the same activations. Their cosine, per layer:

	shallow			deep		
Gemma-9B, stated, RL	+0.976	+0.977	+0.951	+0.396	+0.549	+0.532
Qwen-14B, stated, RL	+0.965	+0.985	+0.985	+0.538	+0.532	+0.583
Llama-8B, stated, RL	+0.928	+0.986	+0.775	+0.591	+0.591	+0.561
Mistral-7B, stated, RL	+0.915	+0.969	+0.879	+0.682	+0.648	+0.589
Gemma-9B, <i>inferred</i> , instr.	+0.732	+0.752	+0.683	+0.620	+0.581	+0.529
Qwen-14B, <i>inferred</i> , instr.	+0.651	+0.708	+0.748	+0.611	+0.420	+0.435

Layers: Gemma 4/8/16 $\rightarrow$ 24/32/42; Qwen 4/8/16 $\rightarrow$ 32/40/48; Llama and Mistral 4/8/16 $\rightarrow$ 20/24/32.

The two fits on the stated-bit task. They recover nearly the same vector in the shallow stack and separate sharply at depth, in all four families, and always at 57–67% of the stack, so the effect is not tied to an absolute layer index. Where the labels are collinear, how you fit does not matter; where they decorrelate, it decides which axis you get.

One thing the cosine is not. The deep value varies only over 0.53–0.60 across these four arms, whose rival deception rates span 0.47 to 1.000. So the angle separates the two fits everywhere at depth, but it does *not* grade with how much the policy’s behaviour aliases truth with action. It is a qualitative signature of the two fits diverging, not a measurement of aliasing.

The last two rows are the informative ones. When the bit is *inferred* rather than stated, truth, action and token are less collinear from the outset, and the shallow cosine is correspondingly lower, at 0.68–0.75 on Gemma and 0.65–0.75 on Qwen, against 0.78–0.99 for every stated-bit row. That is the direction the aliasing account predicts, and it now holds on two families. The contrast is not an arm effect, which the table alone would leave open, since every stated-bit row above is a reward-trained arm and both inferred-bit rows are instructed: the *instructed* stated-bit arm swept in §K.3 sits at 0.950/0.996/0.986 shallow, with the stated-bit rows and not with the inferred ones, so it is the task and not the training route that moves the shallow cosine. We still put it no more strongly than a qualitative pattern: it is two task-contrast rows against four stated-bit rows, the deep values overlap across conditions, and a cosine between coefficient vectors is not itself a measurement of aliasing.

It also settles an ambiguity the AUROCs alone cannot. A separation index of ≈ 2 is consistent both with two distinct directions and with *one* axis whose sign the mixed fit merely resolved. The latter would give $\cos \approx -1$. We measure $\cos \approx +0.55$ at layer 32, so these are two different directions, roughly 57° apart.

K.2 The signed test, and what the ally-fit direction does

We swept α *through zero*, ten values from -1.37 to $+1.37$ times the median residual norm, against a single in-run $\alpha=0$ baseline, so the whole dose–response is one measurement rather than a comparison across runs. On the instructed arm (the one with no floor) the mixed-fit direction gives a monotone, ordered response on true-bit-1 trials over $\alpha_{\text{rel}} \in [-1.37, +0.69]$: 0.365 at the negative end, rising through the 0.419 baseline, to 0.524 at $+0.69$. It then falls back to 0.431 at $+1.37$, which we read as the usual large- α degradation rather than part of the response. So $-\alpha d$ does lower the true-bit-1 rate, and the axis has signed influence: the test we previously flagged as unrun.

Why this does not license a truth-steering reading. The true-bit-0 rate is exactly 0.000 at every α , in both directions, out to $1.37\times$ the residual norm. A direction carrying truth causally should rescue true-bit-0 trials when pushed the other way; it never does. The effect is confined to the subpopulation already near the decision boundary, and it moves in the direction of one token.

The paired, per-trial form. Every α cell runs on the same episodes, so this can be said per trial, not per rate. Matching trials across cells (an ordering we verify by checking that the true bit, codebook and role agree position-by-position across all 1000 records) and applying McNemar to the discordant pairs: on true-bit-1 trials the flips are almost perfectly one-sided: for the ally-fit direction at $\alpha_{\text{rel}} = \mp 0.686$, 351 trials switch toward the truthful answer and 0 switch away ($p \approx 4 \times 10^{-106}$), with the same one-sidedness at every smaller dose and for the mixed-fit direction at smaller magnitude (79 versus 0). On true-bit-0 trials the discordance is exactly zero in 25 of the comparisons: not one trial of 496 changes state, in either direction, at any dose up to 0.686. That closes a loophole the aggregate rates leave open: a 0.000-versus-0.000 comparison is consistent with equal numbers of trials flipping each way and cancelling, and here nothing flips at all. The single exception is the ally-fit direction at the extreme ∓ 1.373 , which rescues 57 of 496 true-bit-0 trials, strictly in one direction ($p \approx 1 \times 10^{-17}$), the large-dose token-preference effect noted below, and nowhere else.

The same question where there is no floor, on two families. Every argument above rests on true-bit-0 sitting at 0.000, and a rate pinned at an extreme cannot fall. The inferred-bit arms are the cells in which *both* rates are interior at baseline (Gemma-9B at true-bit-1 0.240 / true-bit-0 **0.700**, Qwen-14B at 0.609 / **0.145**), so both are free to move in either direction, and the truth-versus-answer question becomes directly decidable. A truth axis pushed toward truth must raise *both* rates; an answer axis must move them in opposition. On Gemma the ally-fit direction moves them in opposition at every one of the nine doses at which it moves them at all, monotonically and symmetrically:

α_{rel}	−1.40	−0.70	−0.35	0	+0.35	+0.70	+1.40
true-bit-1 rate	0.215	0.209	0.224	0.240	0.261	0.279	0.286
true-bit-0 rate	0.766	0.725	0.710	0.700	0.683	0.673	0.692

Pushed one way it makes the model likelier to answer truthfully when the truth is 1 and *less* likely when the truth is 0; pushed the other way, exactly the reverse. That is one token-preference knob, read as “truth” on one subpopulation and “lying” on the other.

Replication on a second family, with a larger effect. We ran the identical contrast on Qwen2.5-14B’s inferred-bit arm, at layer 40 of 48, chosen because that is where *its* ally-fit probe is most action-aligned, reading 0.285, whereas Gemma’s inferred ally-fit probe never inverts. Of the twelve non-zero doses, ten move the two rates in opposition, and over the well-behaved range $|\alpha_{\text{rel}}| \leq 0.535$ the true-bit-0 rate falls *strictly* monotonically ($0.186 \rightarrow 0.114$) while true-bit-1 rises ($0.526 \rightarrow 0.689$), swings of 84 and 35 trials, roughly double what the Gemma cell gives. The two exceptions are a single-trial wiggle at the smallest positive dose and the largest dose $|\alpha_{\text{rel}}| = 1.07$, where both rates degrade together in the representation-damage pattern reported throughout.

The conclusion does not depend on the floor: on both arms where the rates could move freely, the axis fails the truth-axis test in the direction the identification account predicts. The mixed-fit direction is by comparison inert on both ($|\Delta| \leq 0.019$ on Gemma, ≤ 0.041 on Qwen, with no consistent sign and a U-shaped rather than ordered response on Qwen), which is the same ordering seen everywhere else.

The comparison with the ally-fit direction, the unidentified one this paper criticizes. Both directions come from the same fit on the same activations, differing only in which contexts the fit saw, so we can perturb each at the same layer, arm and episodes. We compare them at *matched* α , because the total swing saturates and then hides the difference:

	$ \alpha_{\text{rel}} $	mixed-fit	ally-fit	ratio
Gemma-9B L32, instructed	0.34	+0.125	+0.452	3.6×
Gemma-9B L24, instructed	0.46	+0.427	+0.984	2.3×
Qwen-14B L32, reward-trained	0.27	0.000	+0.036	n/a
Gemma-9B L32, <i>inferred</i> bit	0.35	−0.004	+0.037	n/a

The comparison is at the nearest available *relative* dose, not a matched raw α : the median residual norm differs by layer and model, so the same α is a three times larger perturbation on Qwen than on Gemma. At matched relative dose the ally-fit direction produces the larger displacement in all four cells. We report a ratio in only two of them. A ratio is meaningful only when both arms moved the same way by an amount larger than noise, and in two cells the denominator fails that test: the Qwen mixed-fit response is exactly zero at this dose, and the inferred-bit mixed-fit response is *negative*: a -0.004 displacement is 2 trials out of 517, and pointing the wrong way. Dividing by either would manufacture a large multiplier out of an absent effect, so we print a dash. Pushing Qwen harder separates the two directions without needing one: at $\alpha_{\text{rel}} = 1.09$ the ally-fit direction takes the true-bit-1 rate to 1.000 while the mixed-fit direction is still at 0.000, and the mixed-fit direction only reaches 0.569 at twice that dose. So the claim is a consistent *ordering*, not a multiplier: where both arms move, the ratio is 2.3–3.6×, and at saturation it necessarily goes to 1.

The axis at high dose. At layer 24, from $\alpha_{\text{rel}} = \pm 0.92$ upward, *both* directions achieve total control of the answer: the true-bit-1 rate goes to 0.000 under negative α and 1.000 under positive, with true-bit-0 moving in exact opposition (1.000 and 0.000). The model emits whichever token we choose and the “truth rate” follows mechanically. That is an answer axis, not a truth axis, and it is why the two directions tie at layer 24’s high doses.

Both directions on the inferred-bit task. In absolute terms they become nearly inert. The largest true-bit-1 range we can produce anywhere on the sweep is 0.077 (ally-fit) and 0.021 (mixed-fit), against 0.768 on the stated-bit task at the same layer. The more informative difference is not the magnitude but whether the response is *ordered* in α at all: the ally-fit direction still is, rising monotonically across the sweep (108 $\rightarrow$ 148 truthful trials of 517 from $\alpha_{\text{rel}} = -0.69$ to +1.40), whereas the mixed-fit direction falls weakly with α (134 at the negative end down to 123, before rebounding to 129 at the largest dose) and never leaves a 0.021 band. So on this task the mixed-fit direction has no dose–response to speak of, rather than a small one: it is neither ordered in α nor large enough to be worth a ratio. This is consistent with an axis that controls which token is emitted having little to grip when the target quantity is not that token. That arm also has a different baseline (0.240 against 0.419), a separately trained adapter and a harder task, any of which could reduce sensitivity, so the comparison is consistent with the token account without isolating it from those alternatives.

Why this confirms the diagnosis. On ally data truth, action and emitted token share a label vector, so the ally-fit “truth” probe is an action probe; steering along the axis that predicts the emitted token moves the emitted token. The mixed-fit direction isolates truth *as distinct from* action, and that component has markedly less purchase on the answer. One thing we can *not* claim: the gap does not

visibly grade with how often the policy lies. We see the same ordering at rival deception 0.545 and 0.639 as at 1.000, and the geometric analogue, the deep cosine of §K.1, is likewise flat across that range. Aliasing on the *fitting* data is a property of the protocol rather than of the policy, so there is no reason to have expected a gradient in the deception rate; we report the absence of one so that it is not read into the results later. Identification and causal potency come apart: the better measurement of truth is the weaker lever on behaviour, and what the field has been steering is largely the action axis wearing a truth label.

K.3 The advantage appears exactly where the two directions separate

The comparison above is at one layer, which leaves open whether the ally-fit advantage has anything to do with the two directions *differing*. §K.1 says they are nearly the same vector shallow and separate at depth, so that account makes a prediction: the advantage should be absent where $\cos \approx 1$ and present where the fits diverge. We ran the identical contrast at five layers, sweeping α over a $128\times$ range at each because the median residual norm grows more than tenfold across the stack (83 at layer 8 to 860 at layer 40) and a grid fixed in raw α would not be comparable. The five dose–response pairs are plotted in Fig. 15.

layer	8	16	24	32	40
$\cos(\hat{d}_{\text{ally}}, \hat{d}_{\text{mixed}})$	0.996	0.986	0.773	0.603	0.540
ally-fit advantage	none	none	clear	clear	<i>both inert</i>

Where the two fits are nearly one vector. At layers 8 and 16, $\cos \geq 0.986$ and the two dose–response curves lie on top of each other (Fig. 15, first two panels), with no advantage either way, so the two are causally interchangeable there. Where they separate, at layers 24 and 32, it is the ally-fit direction that carries the behaviour: at layer 32 it drives the true-bit-1 rate from 0.16 to 0.93 across the sweep while the mixed-fit direction moves only 0.36 to 0.52. So the cosine predicts where the advantage appears: the criticized direction is potent only in the regime where the two fits have separated.

We report this as an ordering and not as a potency ratio. A ratio here is not a stable quantity: at small doses the mixed-fit displacement is a noise-level denominator, at large doses both arms saturate and any ratio goes to 1 by construction, and at layer 24 there are doses where the denominator points the *opposite* way so the ratio is undefined rather than large. Quoting a multiplier, or worse a multiplier read off one chosen dose, would put a number on the claim that the underlying data do not support, the same failure we flag for the inferred-bit cell in §K.2. The two curves per layer are shown instead, so a reader can see the denominator that a ratio would hide.

Layer 40, the informative exception. Its cosine is the *lowest* we measure (0.540), so a simple “more separation, more advantage” rule predicts the largest effect there. Instead *both* directions are inert: the mixed-fit spread is slightly negative at every dose and the ally-fit spread never exceeds 0.036. The plain explanation is that layer 40 of 42 has almost no downstream computation left for a perturbation to act through, and we cannot separate that from a claim about the direction. The defensible statement is therefore that direction-separation is *necessary* but not sufficient: the advantage needs both that the two fits differ and that the site has causal purchase at all. Reading the effect off a single chosen dose would make the five layers look monotone in the cosine, and they are not.

One consistency note, since the two look like they should conflict. The advantage grades with the cosine *across depth within an arm*, but not with the deception rate *across arms* at fixed depth, where the deep cosine is itself flat. Both are the same statement: what predicts the advantage is how far apart the two fitted directions are, not how often the policy lies.

Ablation on the ally-fit direction is void exactly as before (post-ablation decodability 0.968 against a 0.600 ceiling), so nothing in this section rests on an ablation. The Qwen arm sits on a 0.000 floor, so its negative half is uninformative by construction. And these are genuinely different axes rather than one axis re-signed (§K.1).

The ablation result is not a null; it is uninterpretable. Projecting out the direction leaves behaviour unchanged, but our pre-registered positive control shows the intervention did not remove the information: a freshly fitted probe still recovers the bit at 1.000 from the ablated activations, against a permutation-calibrated chance ceiling of 0.600. The same holds for every rank of an INLP subspace

1205 ablation we could complete under an *underdetermined* fit, up to $k=8$ (Fig. 14). Re-running the
 1206 fit *determined* ($n=4000 > d=3584$) changes this at $k=2$: post-ablation decodability falls to 0.600
 1207 against 0.727 for the underdetermined fit at the same rank, so the underdetermined subspace was
 1208 under-removing by 0.128, exactly the bias its own guard predicts, and the reason we do not report the
 1209 underdetermined curve as a bound. We have now completed the determined ladder at both layers, and
 1210 the result is that no rank at either layer licenses a causal null. At layer 32, post-ablation decodability
 1211 is 1.000, 0.5996, 0.5911 and 0.6077 at $k = 1, 2, 4, 8$; at layer 24 it is 1.000, 0.939, 0.913 and 0.997.
 1212 Behaviour is unchanged throughout (ally truth rate 1.000).

1213 **Two of the eight cells clear the reported ceiling.** That ceiling is $\max$ (a tolerance we chose, the
 1214 label-permutation null), i.e. the *more permissive* of the two, and our tolerance (0.600) is the looser
 1215 one at every cell. Measured against its own permutation null instead, every cell at both layers sits
 1216 above it: at layer 32 the nulls are 0.576, 0.580, 0.576 and 0.581, so even $k=2$ and $k=4$, which
 1217 clear 0.600 by 0.0004 and 0.009, are 0.020 and 0.015 *above* the calibrated band, and $k=8$ fails both
 1218 criteria outright. So the pass/fail language is an artifact of the permissive $\max$, and we drop it. The
 1219 bit remains partly decodable after every ablation we ran, so a flat behavioural rate is never licensed
 1220 here as evidence about causal use.

1221 **What the completed ladder buys.** A quantitative depth contrast, which is weaker than a pass/fail one.
 1222 The layer-32 ablation drives decodability down to 0.59–0.61 while the layer-24 ablation leaves it at
 1223 0.91–1.00, so the bit is markedly more distributed at layer 24, but neither layer yields an interpretable
 1224 null, and the two passing layer-32 cells are *bounded by void cells on both sides* ($k=1$ and $k=8$), so
 1225 they are a middle band rather than a trend. Removal does not keep improving with rank: $k=8$ is
 1226 worse than $k=4$ at layer 32 (0.608 vs 0.591) and worse at layer 24 as well (0.997 vs 0.913). One
 1227 caveat travels with the whole ladder: the positive control refits at $n=600$ against a 3584-dimensional
 1228 residual, which is why its permutation null sits near 0.58 rather than 0.50: the null absorbs that
 1229 overfitting, which is exactly why it, and not our tolerance, is the criterion we now report against.
 1230 A behavioural null is only evidence about causal use if the ablation worked, so we draw no causal
 1231 conclusion from it. What it does show is a fact about dimensionality: the bit is not confined to
 1232 that rank-1 subspace, since removing the direction leaves it fully decodable from the orthogonal
 1233 complement.

1234 So the honest position is that the bit is linearly present and the mixed-fit direction we propose has
 1235 no demonstrated causal traction at this site, while the ally-fit direction does (§K.2); the ablation arm
 1236 is uninterpretable rather than negative. Four limits on reading even that. Whole-residual patching
 1237 *does* flip answers at 1.000 (App. F), so information at this site is used by something, though that
 1238 intervention replaces every role-, prompt-, action- and truth-associated component at once, so it says
 1239 nothing about how many dimensions carry the effect. Our α range initially reached only 8.6% of the
 1240 median residual norm. We extended it sixteenfold, to $1.35\times$ the residual norm, with an in-run $\alpha=0$
 1241 control passing through the same hook on the same episodes. On the saturated arm every cell still
 1242 reads 0.000, since adding a vector larger than the residual itself never moves the answer, so “you did
 1243 not push hard enough” is excluded over that range.

1244 On the instructed arm the extended sweep does produce movement. The rival-truth rate rises from
 1245 a measured baseline of 0.211 to 0.264 at $\alpha_{\text{rel}} = 0.69$ (nominally significant), then *collapses back*
 1246 to 0.217 at 1.37. We pre-registered three criteria for counting a move as steering (monotone in
 1247 α , sign-appropriate under an oracle sign, and not mirrored by the unsigned arm), and it fails all
 1248 three: the curve is non-monotone, and the signed and unsigned arms are identical to four decimals
 1249 (which, as noted above, the construction forces on exactly the trials where the movement occurs). The
 1250 mechanism is visible in the breakdown: on true-bit-0 trials the rate is exactly 0.000 at every α in both
 1251 directions, so the axis never induces a truthful answer there, and all movement is confined to true-bit-1
 1252 trials. That is the signature of a shift in *answer bias*, raising the probability of one particular token,
 1253 which is truthful on one subpopulation and deceptive on the other, not of steering toward truth. The
 1254 decisive follow-up, whether $-\alpha d$ lowers the true-bit-1 rate, is answered affirmatively in §K.2; it does,
 1255 and the axis is nonetheless an answer axis rather than a truth axis, for the reason given there. And this
 1256 is one site in one layer of one model family. Finally, App. F’s “no arm difference” is stated at the
 1257 strength each measure supports, which is not uniform. On the answer-position readout the matched
 1258 difference is an *exact* tie: every one of the 8 matched cells reads 0.000 in both arms, so the paired
 1259 difference has zero variance and a conventional equivalence test is undefined rather than passed;
 1260 what bounds it is saturation, not statistics. On the two matched measures that are *not* saturated, a
 1261 two-one-sided-tests procedure does establish equivalence at a ± 0.05 tolerance (readout direction

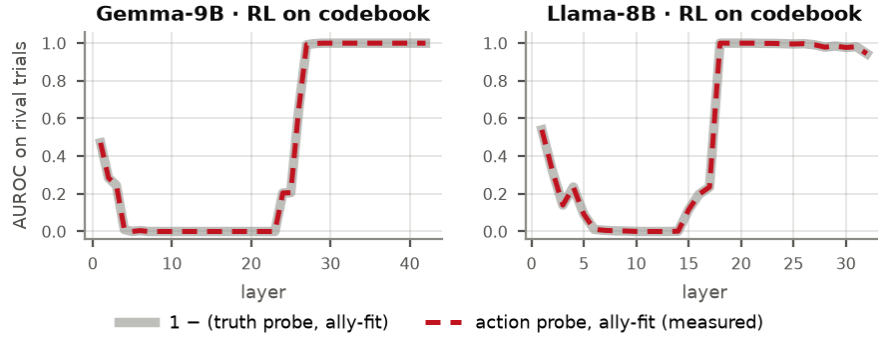


Figure 6: **The identity is forced.** The measured ally-fit *action* probe (dashed) lies exactly on $1 -$ (ally-fit *truth* probe) (thick grey) at every layer, because on ally data the two label vectors are identical and the two “probes” are one fit scored against opposite labels. Maximum deviation over all 751 (cell, layer) pairs: 2.2×10^{-16} .

1262 cosine, mean difference $+0.021$, $p = 0.034$; rival entropy, mean difference -0.003 , $p < 10^{-6}$). On
 1263 the off-position readout it does **not**, for the reason given in App. F. We prefer reporting this unevenly
 1264 to reporting a single tolerance that would be true of some measures and false of others.

1265 L Supplementary figures

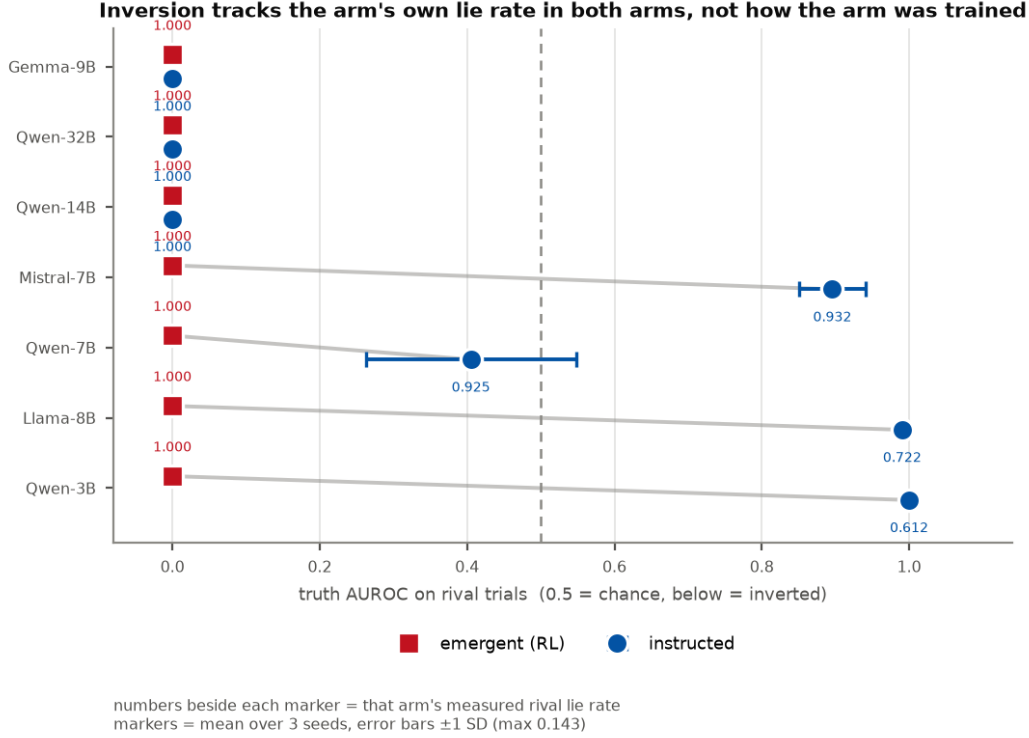


Figure 7: **Being reward-trained is not what produces the inversion.** The behaviourally-matched null of App. F. One row per model family; within a row the **red square** is the reward-trained (emergent) arm and the **blue circle** the instructed arm, dodged vertically so families where both sit at exactly the same AUROC still show two markers. Horizontal position is final-layer truth AUROC on held-out rival trials, so the dashed line at 0.5 is chance and everything left of it is inverted. Markers are the mean over 3 seeds with ± 1 SD error bars, and the number beside each marker is that arm's *measured rival lie rate*, not its AUROC. Rows are ordered by the instructed arm's lie rate, the variable a rate-based account would make the operative one. Read it as follows: every emergent arm lies on 1.000 of rival trials and every one reads 0.000; the three instructed arms that also lie on 1.000 (Gemma-9B, Qwen-32B, Qwen-14B) read 0.000 *too*, while the instructed arms that lie less reliably read well above chance. So how the deception was induced does not predict the readout. *Within this grid* the inverted arms are also the saturated ones, but that relation does not hold generally, and we do not assert it: an arm at rival deception 1.000 that never inverts is reported in App. G. This is a null with tight bounds rather than demonstrated equivalence, and the error bars are 0 in every saturated cell because those readouts sit at an extreme where the measurement is exact.

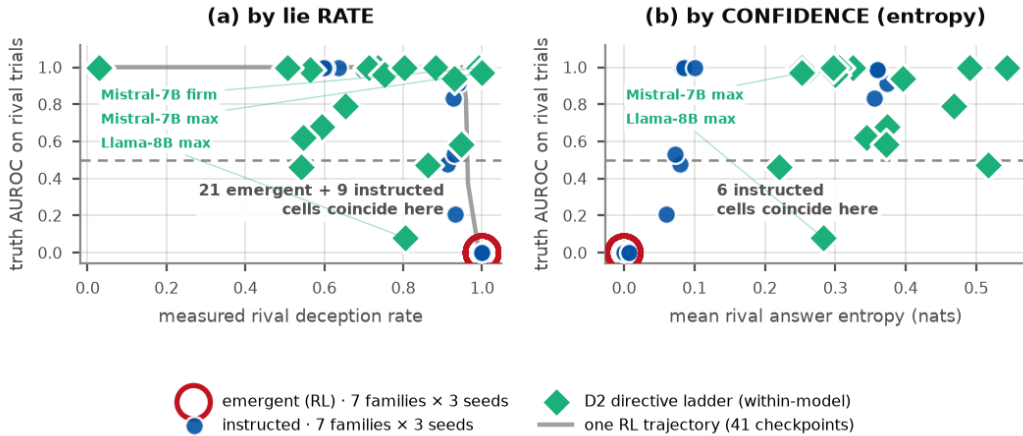


Figure 8: Neither lie rate (a) nor per-example confidence (b) orders the readout: cells with the same behaviour sit at both 0.00 and 1.00. Both were hypotheses of ours, and both are refuted (App. G).

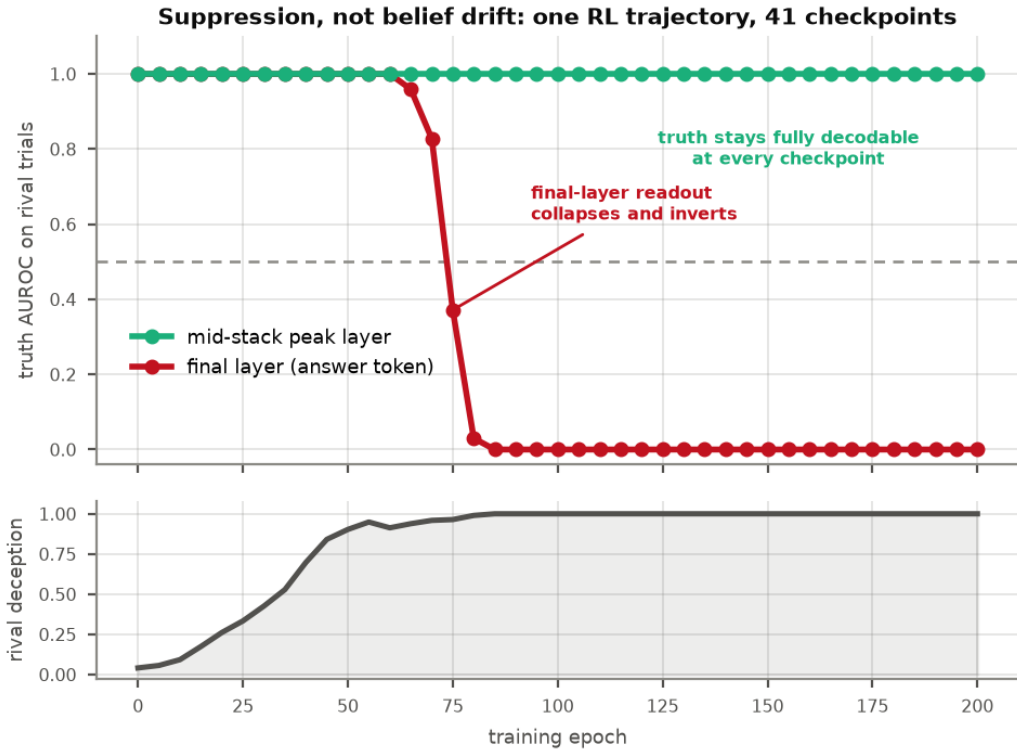


Figure 9: Suppression, not belief drift: one RL trajectory, 41 checkpoints. Mid-stack decodability stays pinned while the final-layer readout collapses and inverts.

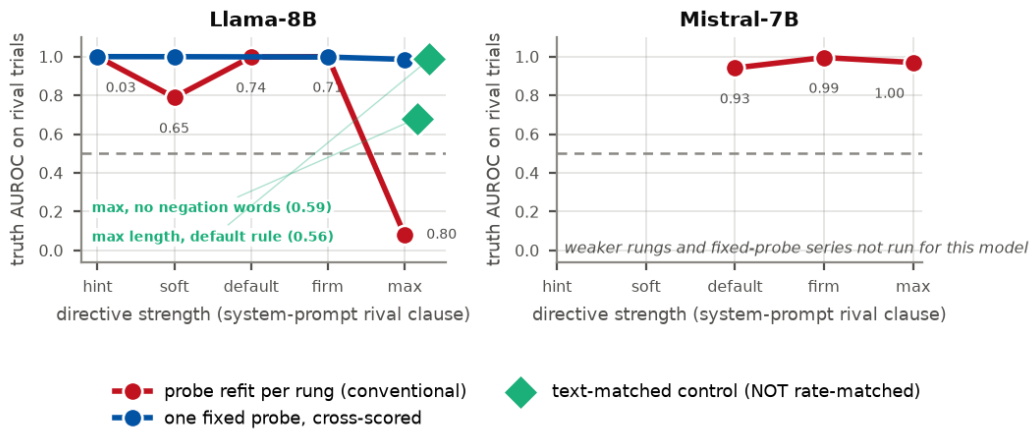


Figure 10: The directive-strength ladder moves the refit probe, not the frozen one.

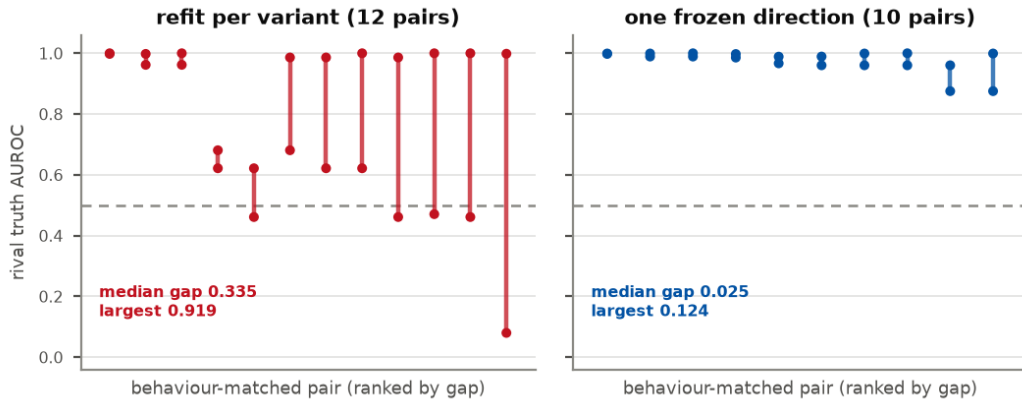


Figure 11: **Behaviour-matched prompt pairs: refitting splits them, one frozen direction does not.** Every pair of the 15 ladder rungs (the reference plus 14 variants) whose rival deception rates agree to within 0.05, each drawn as a segment joining the two members’ rival truth AUROCs, ranked by gap. Left: a probe refit on each variant’s own ally data, median gap 0.335, largest 0.919, and four pairs straddle chance. Right: one direction fit once on the reference variant and cross-scored, median gap 0.025, largest 0.124. Two pairs contain the reference variant itself, which has no frozen score by construction, so the right panel has 10 pairs to the left panel’s 12. Pairs are enumerated under a stated tolerance, not selected on the size of their gap.

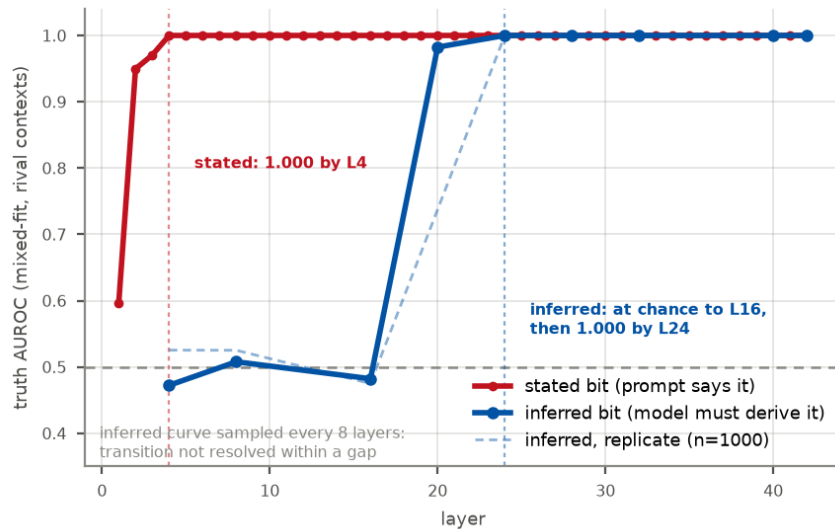


Figure 12: **The depth step.** Truth AUROC of the identified (mixed-fit) probe on held-out rival trials, Gemma-9B. When the prompt *states* the bit (red) it is linearly readable almost immediately and reaches 1.000 by layer 4: reading it is a lexical operation. When the model must *derive* the bit (blue, $N=2000$; faint dashed is the $N=1000$ replicate) the same protocol sits at chance through layer 16 and only then rises, reaching 1.000 by layer 24. The contrast is what licenses treating the deep readout as a computed quantity rather than a copy of the prompt (§7).

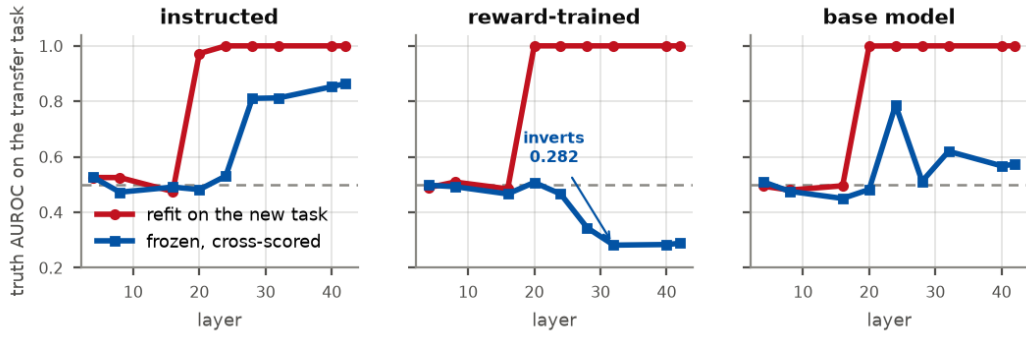


Figure 13: **Refitting certifies almost nothing about transfer.** Each panel scores the inferred-bit task. Red: a probe *refit* on that task, the conventional practice; it reads ≈ 1.000 from layer 20 in all three arms. Blue: the *same* direction fit on the stated-bit task and merely cross-scored. It never exceeds 0.864 on the instructed arm, and on the reward-trained arm it inverts to 0.282. A refit number and a frozen number answer different questions, and only the frozen one bears on transfer.

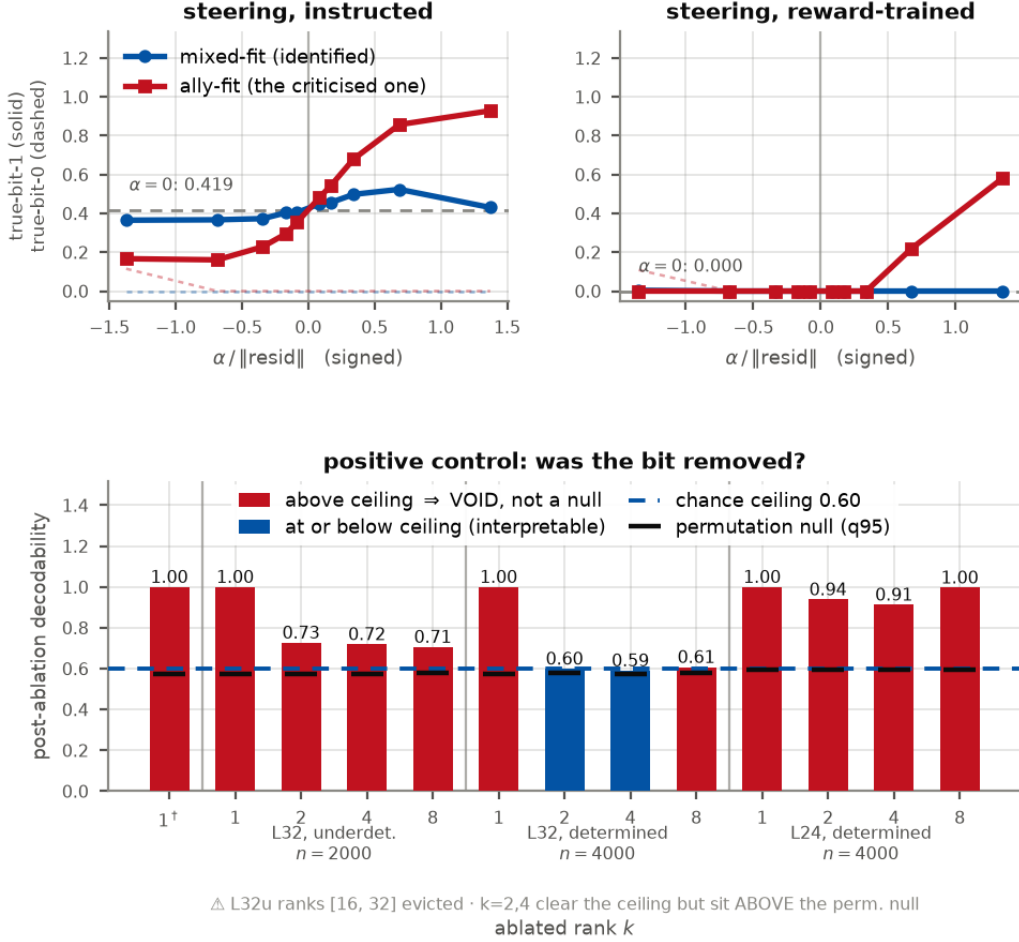


Figure 14: **The criticized ally-fit direction is the causally potent one.** Left and centre: steering at layer 32, swept *through zero* against a single in-run $\alpha=0$ baseline. Solid lines are the true-bit-1 rate, dashed the true-bit-0 rate. The mixed-fit direction (blue, the identified one) responds weakly, $0.365 \rightarrow 0.524$ before degrading at the largest dose; the ally-fit direction (red, the one we criticize) swings $0.167 \rightarrow 0.929$, and on the saturated arm breaks a floor the mixed-fit direction never moves. True-bit-0 stays pinned at 0.000 almost everywhere, lifting only under the strongest negative ally-fit push: this is a token-preference axis, not a truth lever. Right: the pre-registered positive control. Post-ablation decodability must fall *below* the ceiling (0.600) for a behavioural null to be interpretable; red bars fail and are void. Both determined ladders are now complete, and no cell at either layer falls below its own permutation null (black dashes). Ranks 2 and 4 at layer 32 clear the 0.600 tolerance by 0.0004 and 0.009, but sit 0.020 and 0.015 *above* the calibrated null, and $k=8$ (0.608) fails both, so the two clearing cells are a middle band bounded by void cells at $k=1$ and $k=8$. The contrast with layer 24 is one of degree (0.59–0.61 versus 0.91–1.00), not of pass versus fail.

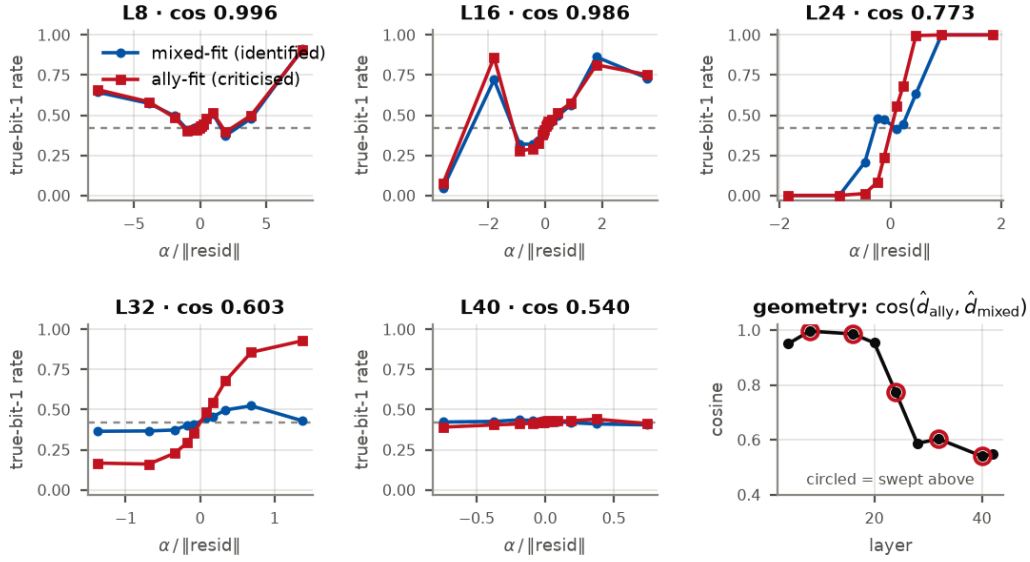


Figure 15: **The ally-fit advantage appears only where the two fitted directions differ, and separation alone is not enough.** Each panel is the same causal contrast run at one depth on the Gemma-2-9B instructed arm: true-bit-1 rate against dose, mixed-fit direction in blue, the criticized ally-fit direction in red, grey dashed line the in-run $\alpha=0$ baseline. At layers 8 and 16, where $\cos \geq 0.986$, the two curves lie on top of each other: the directions are nearly one vector and are causally interchangeable. At layers 24 and 32 they separate, and it is the ally-fit direction that carries the behaviour ($0.16 \rightarrow 0.93$ at layer 32 while the mixed-fit direction moves $0.36 \rightarrow 0.52$). At layer 24 both saturate to 1.000 at the largest doses, so the apparent tie there is saturation rather than equivalence. Layer 40 is the informative exception: it has the lowest cosine we measure yet both arms are flat, so direction separation is *necessary but not sufficient*: with two layers of downstream computation left there is nothing for a perturbation to act through. Bottom right: the geometry the five panels are keyed to, with the swept layers circled. *We plot both dose-response curves rather than their ratio*: at small doses the mixed-fit displacement is a noise-level denominator and at large doses both arms saturate, so a ratio is undefined at one end and 1 by construction at the other. Note the horizontal scales differ by panel: the median residual norm grows from 83 at layer 8 to 860 at layer 40, so a fixed α grid spans a much wider $\alpha/\|\text{resid}\|$ range shallow, and the outermost shallow doses are representation damage rather than dose-response.

1266 NeurIPS Paper Checklist

1267 1. Claims

1268 Question: Do the main claims made in the abstract and introduction accurately reflect the
1269 paper’s contributions and scope?

1270 Answer: [Yes]

1271 Justification: The abstract and §3 state both the corrective claim (ally-fit “truth” probes are
1272 unidentified) and the constructive one (truth/mixed returns to 1.000), and both are scoped
1273 in place: the headline conjunction of a saturated policy *and* a wholly unmodified recipe is
1274 stated as one family (§3.3), the truth/mixed result as 19 of 21 identification cells with
1275 both exceptions named (the separate set of 21 emergent grid cells that carries the saturation
1276 nulls is a different denominator and is labelled as such), and the causal claim scoped to
1277 the axis and the outcome actually tested: the criticized ally-fit direction is the *potent* lever
1278 and the identified mixed-fit direction largely inert (App. K, §K.2), established by a paired
1279 McNemar test on matched episodes rather than offered as suggestive. The scope of that
1280 claim is answer choice, not truth: on the two arms where neither rate sits on a floor the
1281 same push raises the true-bit-1 answer rate and *lowers* the true-bit-0 one, which is what the
1282 paper concludes it is, one token-preference knob rather than a truth axis, so no truth-steering
1283 capability is claimed anywhere.

1284 2. Limitations

1285 Question: Does the paper discuss the limitations of the work performed by the authors?

1286 Answer: [Yes]

1287 Justification: §7 is a dedicated section covering the single one-token game, the one-of-
1288 four-families saturation scope, the floored subpopulations, single-layer and single-site
1289 steering, the settling experiment’s imperfect and strongly depth-dependent transfer, and the
1290 crossed text-versus-behaviour design that this probe site cannot support. Negative and non-
1291 replicating results are *summarised in the body*, inside that same section, with the per-result
1292 detail in App. G: the body names them, so a reader who stops at page nine still sees that two
1293 of our own hypotheses were refuted, and the appendix carries the evidence rather than the
1294 disclosure.

1295 3. Theory assumptions and proofs

1296 Question: For each theoretical result, does the paper provide the full set of assumptions and
1297 a complete (and correct) proof?

1298 Answer: [Yes]

1299 Justification: The two formal results are stated with their assumptions and derived in
1300 place rather than in numbered theorem environments: the forced identity $\text{action/ally} = 1 - \text{truth/ally}$
1301 follows from the ally-data label vectors for truth and action being
1302 identical plus the antisymmetry of AUROC under label negation (§3), and is additionally
1303 verified numerically at all 751 (cell, layer) pairs to a maximum deviation of 2.2×10^{-16} ;
1304 the aliasing audit γ (Eq. 2) is derived in one line in App. D under the stated honest-rung
1305 assumption.

1306 4. Experimental result reproducibility

1307 Question: Does the paper fully disclose all the information needed to reproduce the main ex-
1308 perimental results of the paper to the extent that it affects the main claims and/or conclusions
1309 of the paper (regardless of whether the code and data are provided or not)?

1310 Answer: [Yes]

1311 Justification: §2 gives the game and reward structure (REINFORCE; ally +1/ − 1, rival
1312 +1.5/ − 2), the adapter configuration ($r=16$, $\alpha=32$, on q/k/v/o), the exact base-model
1313 versions, dtype, the transformers version, N per cell, the read position, the three label
1314 sets, the two fitting regimes and the held-out ally / held-out rival scoring split. Where a
1315 number depends on a choice we made (chance tolerance, permutation count, α normalisation)
1316 the choice is stated with the number.

1317 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: We release the game, the randomized-codebook variant, the RL recipe, all probe-fitting regimes, the intervention code and the figure pipeline, together with the exact per-run commands for every cell reported here (§8). For review the code is provided anonymised at the repository link given in §8. The figures in this paper are rendered from a consolidated results file that ships with the code, so every plotted number is re-derivable without re-running a GPU job. We deliberately do *not* claim a one-command hosted reproduction path: reproducing the underlying runs requires an accelerator of the class named in §2, and we state the commands rather than ship a wrapper we have not executed end to end. Independently of the release, the paper states the protocol in enough detail to reproduce the main results from scratch (question 4).

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Measurement-side settings are fully specified in the paper. Probe type (per-layer logistic), read position (final prompt position, the answer-prediction position), layer grids, N per cell, the ally-only versus mixed ally+rival fitting regimes and the held-out scoring split are in §2; the intervention settings (rank ladders, fit sizes relative to the residual width, dose grids in α_{rel} rather than raw α) are given where used in §K.2 and §K.3. Since every claim in the paper is a claim about a *probe* or an *intervention*, these are the settings the reported numbers are a function of. Training-side settings are split between the paper and the release, and we state which is which. §2 gives the RL algorithm (REINFORCE), the reward values, the adapter configuration ($r=16$, $\alpha=32$, on q/k/v/o) and a qualitative stopping rule (“until reward plateaus”); the gradient optimizer, learning rate, batch size and epoch budget are *not* in the paper and are in the released per-run commands (question 5), which is where a replicator obtains them. One gap is visible in the paper and we flag it rather than let a reader assume the text is complete: Table 1 has a row named “larger step size / batch” whose two values appear nowhere in the text. That row is a robustness arm rather than a load-bearing cell, being one of five recipes that all give the same contrast, so the omission does not affect a reported conclusion.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Where intervals exist they are episode-level bootstraps; ablations are judged against *permutation-calibrated* nulls rather than a fixed band alone, and cells clearing the tolerance but sitting above their permutation null are reported as suggestive, not established (§K.2). We also state the limits of our own error bars: the bootstrap holds the fitted probe and the partition fixed, so it is not a run-to-run interval, and we quantify refit variance separately across five split seeds in App. J (a new held-out wording set and a new episode partition each time), where the across-refit sd is small at some depths and an order of magnitude larger at others, which is why the layer a claim is anchored at is stated rather than left implicit; §4, by contrast, reports a resample of the ally *subset* and is a different quantity; one equivalence test is reported as *undefined* because its paired difference has exactly zero variance, and single-run cells without intervals are named as such.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: §2 states the type of compute worker (a single H200-class GPU per run), dtype (bf16), the software version and the per-cell N , which fixes the scale of every reported cell and is what a replicator needs in order to know what hardware to obtain. Two of the three things the question names are not reported, and we state that rather than leave it inferred: we give no GPU memory footprint and no wall-clock time for any experiment, and no aggregate, neither total GPU-hours nor the cost of the preliminary and failed runs, of which there were many, so the figure a replicator would budget is larger than the sum of the cells in the paper and we cannot say by how much. The two omissions are of different weight. Memory is bounded above by the accelerator named, so the reported worker type already constrains it; time of execution was not instrumented, and we will not report a figure we did not measure.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: We have reviewed the NeurIPS Code of Ethics and are not aware of any respect in which this work departs from it. No human subjects, no personal data, and no scraped content are involved.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The paper’s central result is itself the safety-relevant impact and is discussed as such: a probing protocol already proposed for deception monitoring can return confidently inverted readings that in-distribution validation cannot detect (§3, App. D), so the negative impact of the *status quo* is false confidence in a deployed monitor. The positive side is the constructive fix and five prescriptions. We scope the contribution to the identifiability of the measurement rather than to detector deployment (§7), which is the claim that could otherwise be over-read.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The artifacts pose no realistic misuse risk requiring safeguards: the deceptive policies are adapters trained on a synthetic one-bit guessing game with a single-token answer, and they confer no capability that transfers outside it. No trained artifact is released: no model or adapter weights, no fitted probe or steering vector, and no corpus of model-generated text (the results file that does ship, question 13, holds aggregate measurements such as AUROCs and rates, not generations), so nothing ships that could be repurposed as a deceptive policy or as an off-the-shelf detector. What does ship is *code* (questions 5 and 13), including the probe-fitting and intervention code; running it requires supplying one’s own model and one’s own compute, and it produces a measurement rather than a capability.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Every base model is credited by exact name and version in §2, and App. A.2 names the governing license for each one individually, noting that they are not uniform: Mistral-7B-v0.3 and Qwen2.5-{7,14,32}B are Apache-2.0, whereas Llama-3.1-8B is under the Llama 3.1 Community License, Gemma-2-9B under the Gemma Terms of Use, and Qwen2.5-3B under the Qwen Research License rather than Apache-2.0: Qwen exempts its 3B and 72B variants. All use here is non-commercial research, which every one of these licenses permits.

1426 **13. New assets**

1427 Question: Are new assets introduced in the paper well documented and is the documentation

1428 provided alongside the assets?

1429 Answer: [\[Yes\]](#)

1430 Justification: There are two new assets, both provided anonymised at the repository link in

1431 §8 (question 5). (i) *Code*: the game and its randomized-codebook variant, the RL recipe, the

1432 probe-fitting and intervention code and the figure pipeline, with the exact per-run command

1433 for every cell reported here. (ii) *Data*: a single consolidated results file holding the aggregate

1434 measurements (AUROCs, deception rates, cosines, dose responses) that every figure and

1435 table in this paper is rendered from, so a reader can re-derive any plotted number without a

1436 GPU. Documentation accompanies both: a description of the package layout, the per-run

1437 commands, and, for the results file specifically, the figure pipeline that consumes it ships

1438 alongside it, so the mapping from a stored field to a printed number is executable rather

1439 than merely described. The terms under which both assets are released are declared in the

1440 package itself, where a user will look for them, and no third-party asset is redistributed

1441 inside it: no base-model files, no weights and no adapters, so ours are the only terms that

1442 need declaring. **No trained weights or adapters are released**, so the redistribution and

1443 naming obligations that App. A.2 records for the Llama and Gemma terms do not attach to

1444 this release.

1445 **14. Crowdsourcing and research with human subjects**

1446 Question: For crowdsourcing experiments and research with human subjects, does the paper

1447 include the full text of instructions given to participants and screenshots, if applicable, as

1448 well as details about compensation (if any)?

1449 Answer: [\[N/A\]](#)

1450 Justification: The work involves no crowdsourcing and no human subjects. All data are

1451 generated by the synthetic game described in §2.

1452 **15. Institutional review board (IRB) approvals or equivalent for research with human**

1453 **subjects**

1454 Question: Does the paper describe potential risks incurred by study participants, whether

1455 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

1456 approvals (or an equivalent approval/review based on the requirements of your country or

1457 institution) were obtained?

1458 Answer: [\[N/A\]](#)

1459 Justification: No research with human subjects was conducted, so no IRB or equivalent

1460 review was required.

1461 **16. Declaration of LLM usage**

1462 Question: Does the paper describe the usage of LLMs if it is an important, original, or

1463 non-standard component of the core methods in this research? Note that if the LLM is used

1464 only for writing, editing, or formatting purposes and does *not* impact the core methodology,

1465 scientific rigor, or originality of the research, declaration is not required.

1466 Answer: [\[N/A\]](#)

1467 Justification: LLMs are the *object of study* here, not a component of the method: the models

1468 named in §2 are what we train and probe. No LLM plays an important, original or non-

1469 standard role in the core methodology, which is probe fitting, activation interventions and

1470 the statistical tests reported above.